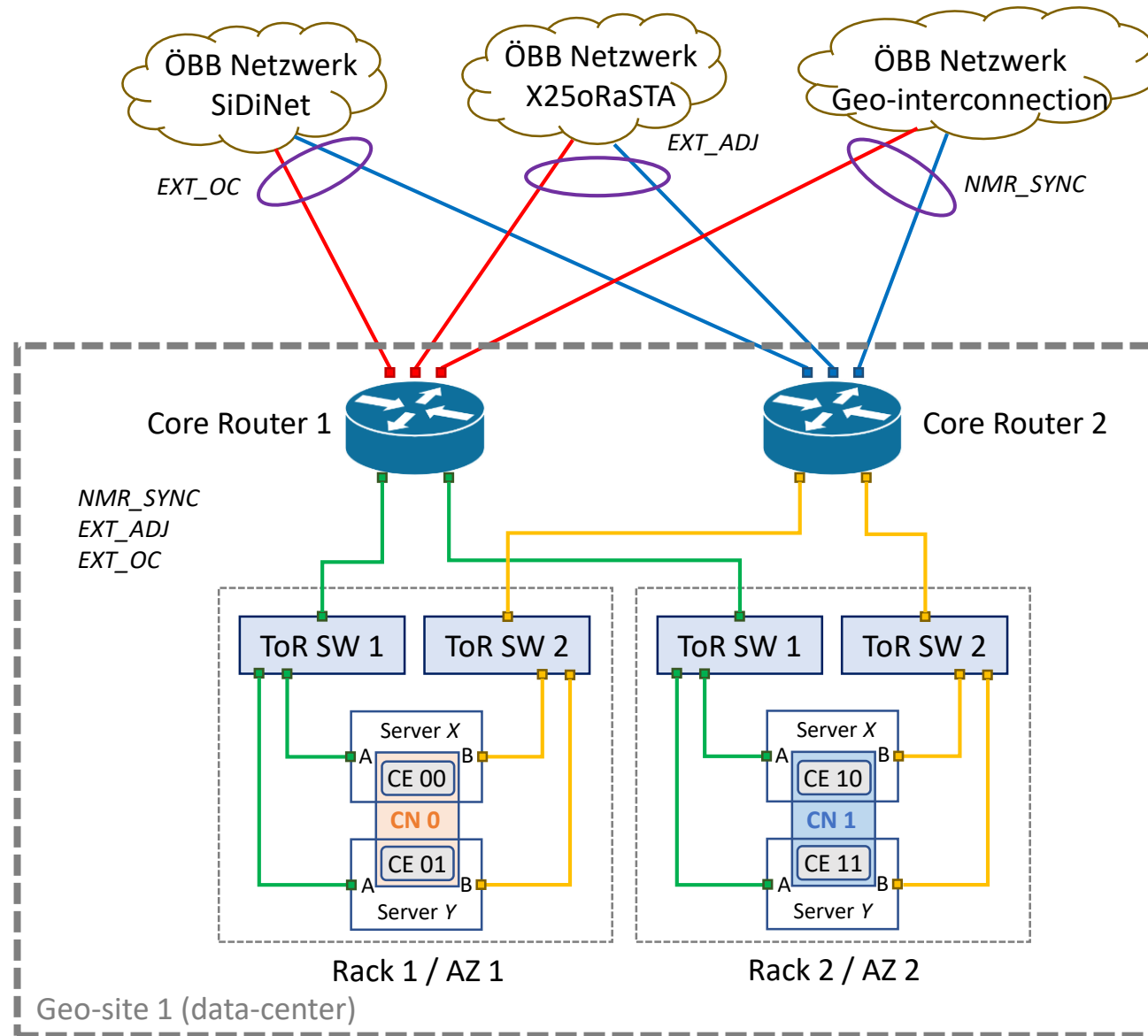


Network topology for the initial split-brain analysis in CG



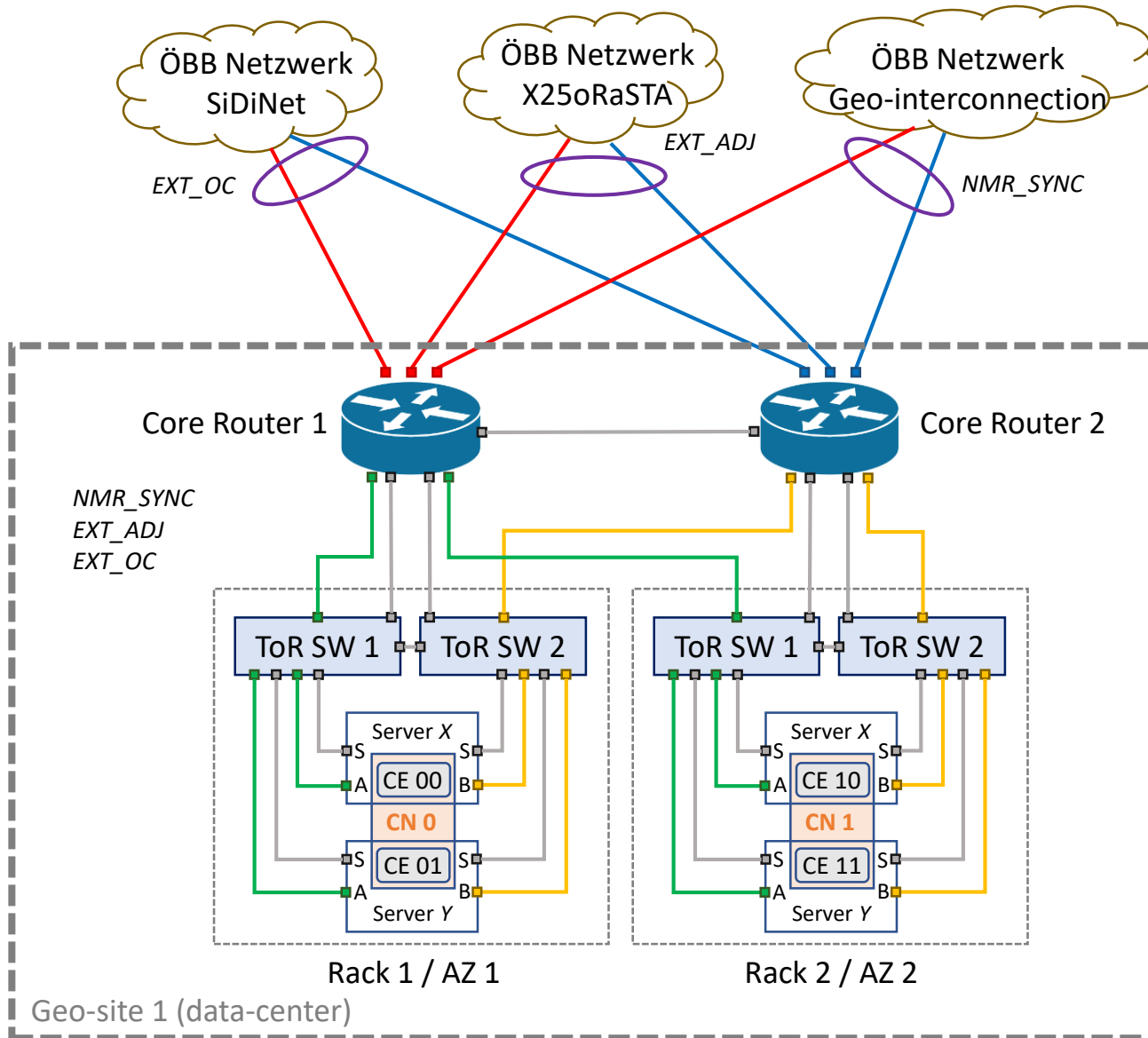
Assumptions:

- ECMUX is not considered in this initial analysis
- One CN is inside one availability zone (rack)
- CEs of the same CN are deployed on different server blades
- There is at most one switch between two CEs of CN (SDD NMR and network requirements)
- "IPS" Network for other interfaces (SECLOG, SYSLOG, SECMNT, DLG ...) not relevant for this initial analysis, thus not shown and considered
- Bonded interface(s) for IPS control plane not relevant for this initial analysis, thus not shown and considered
- Single point of failure between communication points is excluded due to the network topology design
- Double failures are considered

Notation:

- *Italics* denotes VIL's virtual networks assigned to the physical network
- Ports A and B on the server blade are each from different network card, thus failure of one card affects only one channel and not both. The other 2 ports are not relevant for this initial analysis and model.

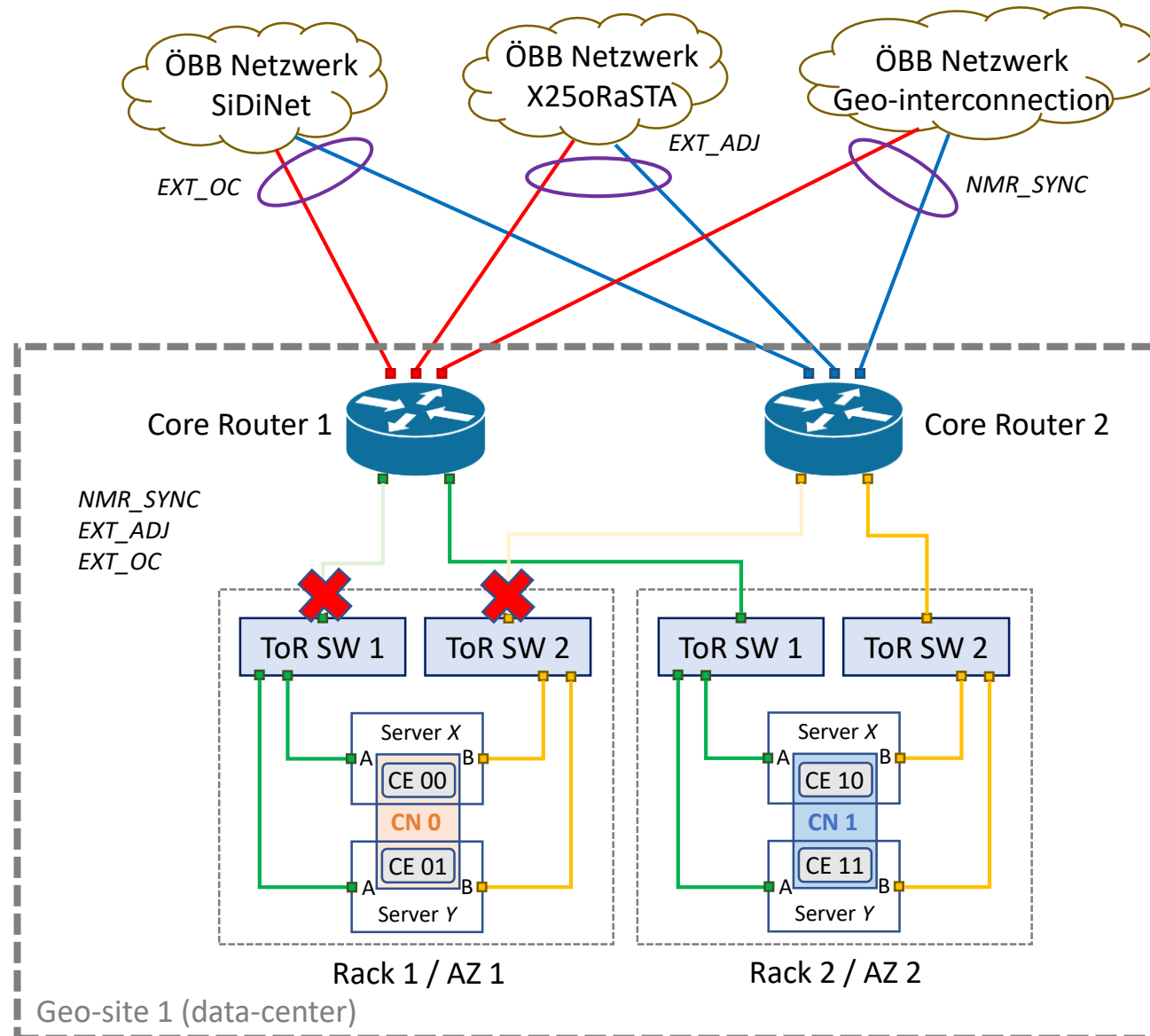
Network topology for split-brain analysis 2x ToR SW with bonded service channel



Assumptions:

- 2 NIC ports for independent physical links
- Other 2 NIC ports form bonded logical link (grey, S) – for control plane, and service (IPS) networks
- TAS Plf Supervision channel (Sv) is deployed on bonded link
- Observer is deployed in AZ where are no CNs of the given CG running
 - Increased availability in case of CG start, quick CN recovery
 - We have 6 AZ (3x2), CG is deployed in 4. That leaves 1 AZ at each geo-site for potential Observer deployment

Double failure between racks and core routers



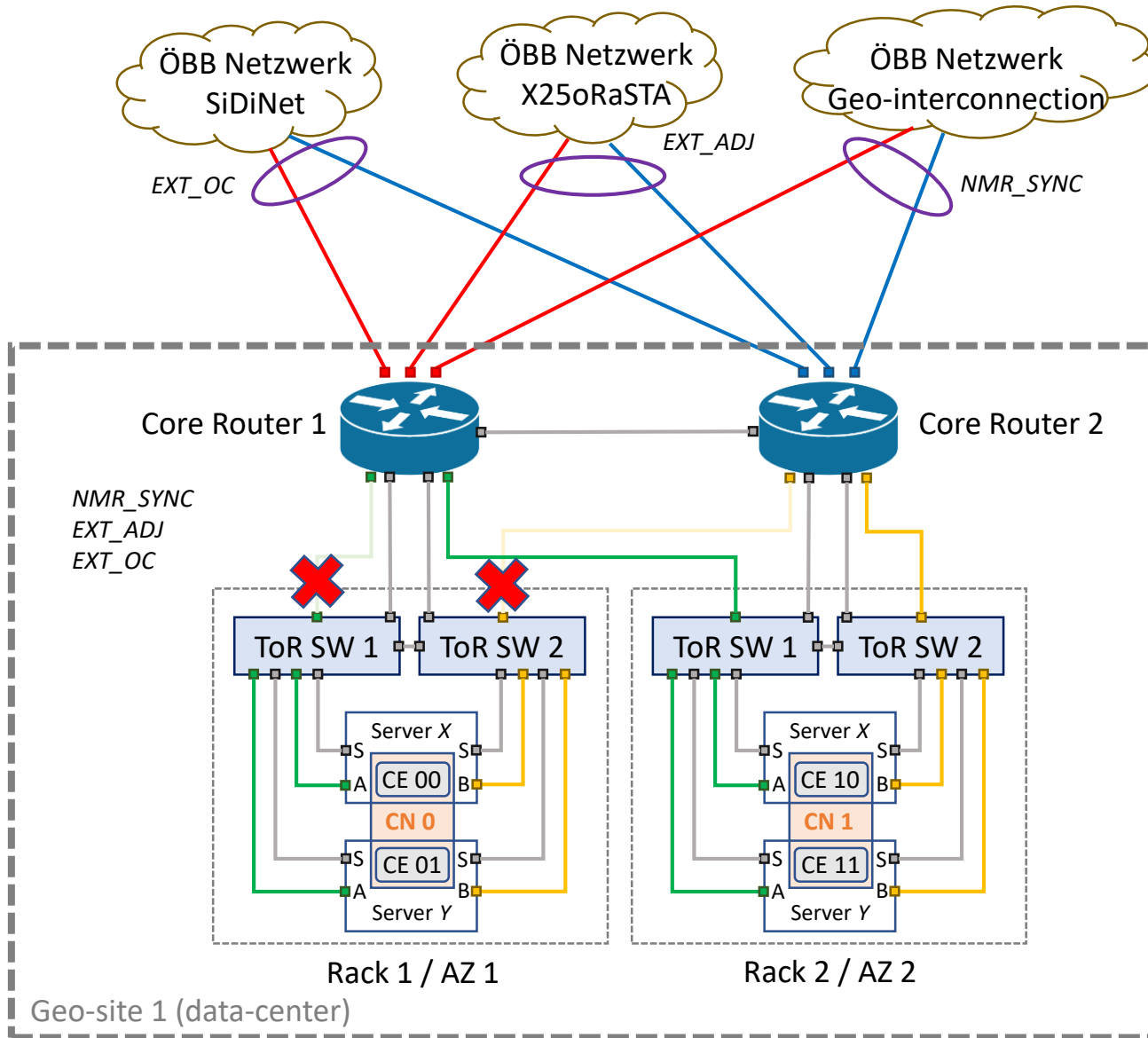
Master CN (on the left):

- **CN effect:** CN is operational but cannot send/receive any communication from CG and external networks. Remains master.
- **Effect on the rest CG:**
 - Master election process (MEP) starts
 - MTP does not elect master
 - Slaves will reset (master election timeout). Without Observer will not be able to elect a new master at startup.
- **External networks:** No communication with Adjacent systems. EXT_OC – tbd
- **No risk of split brain**
 - No safety hazard because CN0 is fully isolated and cannot communicate with external networks
- **Availability issue – state is lost because slaves go to reset. When master is reintegrated – potential loss of state again.**

Slave CN:

- **CN effect:** CN is operational but cannot send/receive any communication from CG and external networks.
- **CG effect (default):**
 - The rest of CG continues normal operation
 - CN0 goes into master election process (MEP)
 - MTP does not elect master
 - Slave will reset. If configured for Observer, then will not elect master and continue in reset (if no MCI configured).
- **External networks:** No effect
- **No risk of split-brain**
 - CN0 can't decide to be the master and continues MEP until the connection is back

Network topology for split-brain analysis 2x ToR SW with bonded service channel



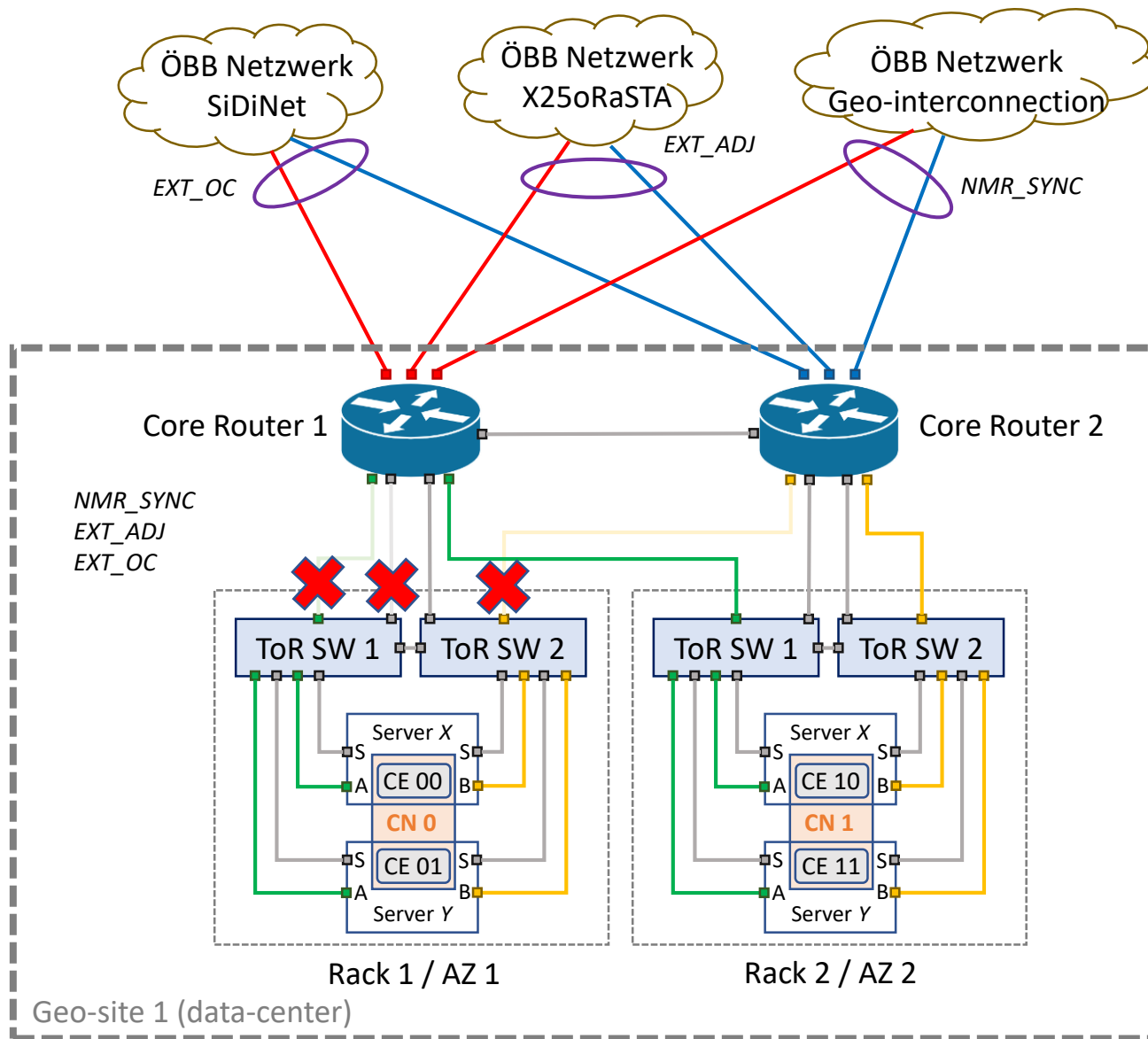
Master CN (on the left):

- CN0 remains master, will not see other slaves. No communication with OCs and ADJ, no advancing of computation and processing of messages from EXT networks.
- CN1-CN3: Master tick timeout → MEP, MTP → Not_OK, Sv → Not_OK, **panic** → New master NOT elected (because Observer sees CN0)
- **Availability issue because no communication with EXT networks on master CN**
- **No hazards, no split-brain**
- **Reintegration – CN0 maintains the state**

Slave CN (on the left):

- **CN0:** Master tick timeout → MEP, MTP → Not_OK, Sv → Not_OK, **panic** → reset, ends in recovery (because Observer sees master CN)
- **CN1-CN3:** Operational, including master CN.
- **No hazards, no split-brain**
- **Reintegration – CN0 recovers**

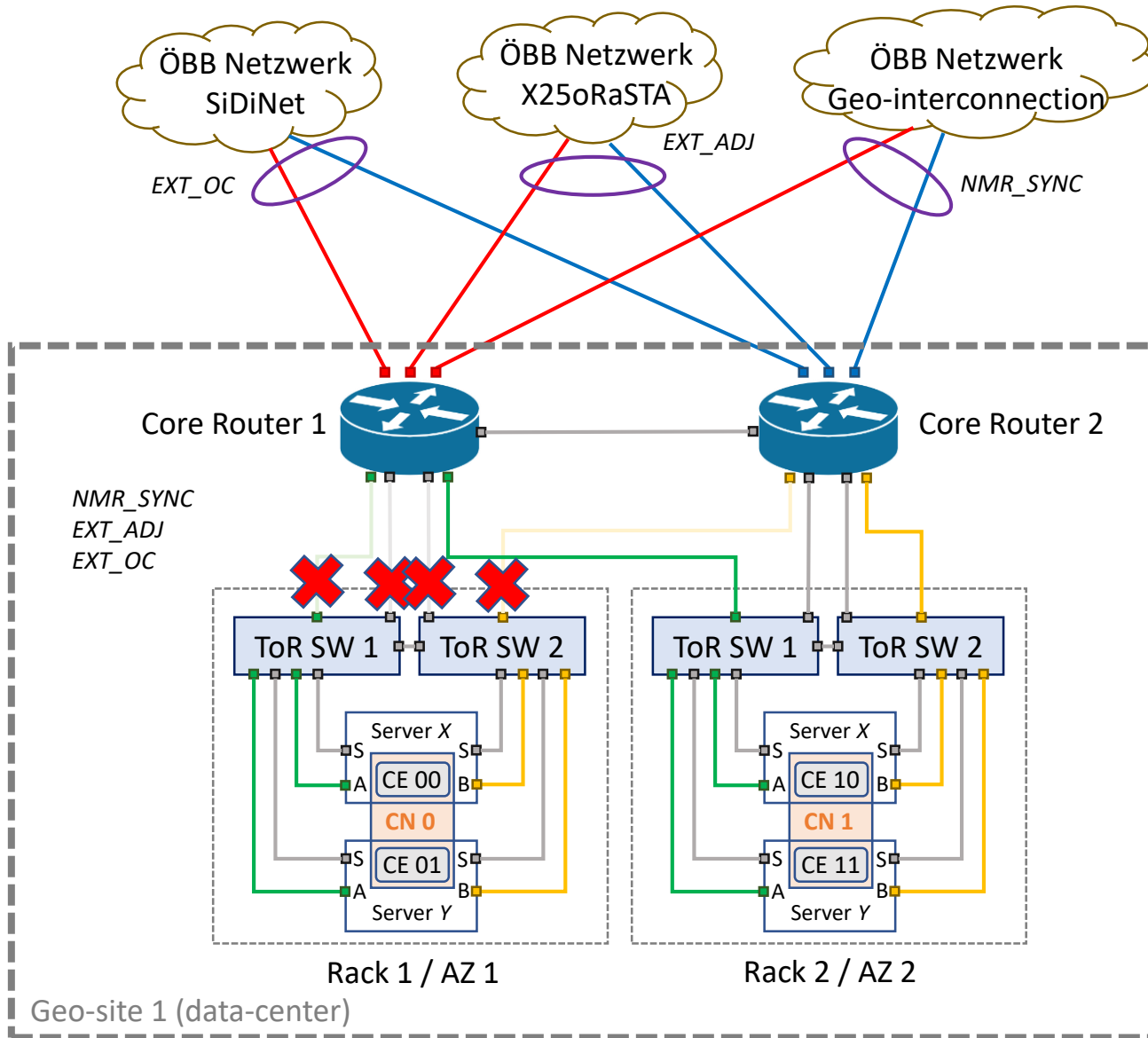
Network topology for split-brain analysis 2x ToR SW with bonded service channel



The same as previous (link fault on both Yellow and Green)

Because Service channel is still operational, i.e. Sv and Observer are reachable from within the Rack1/AZ1.

Network topology for split-brain analysis 2x ToR SW with bonded service channel



Quadruple link failure – out of spec, unlikely

CN0 Master (No MCI):

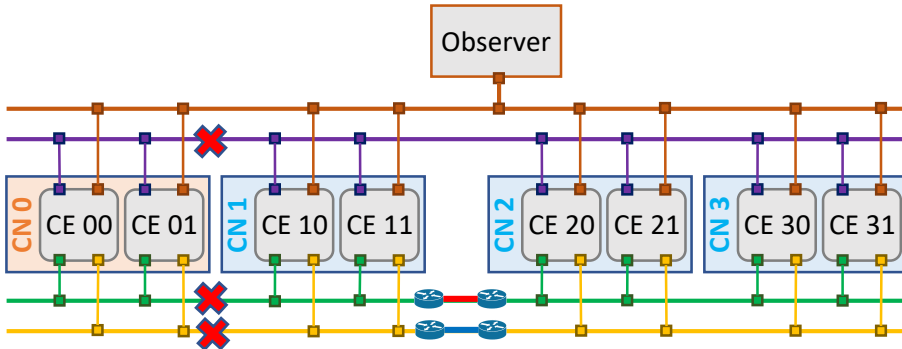
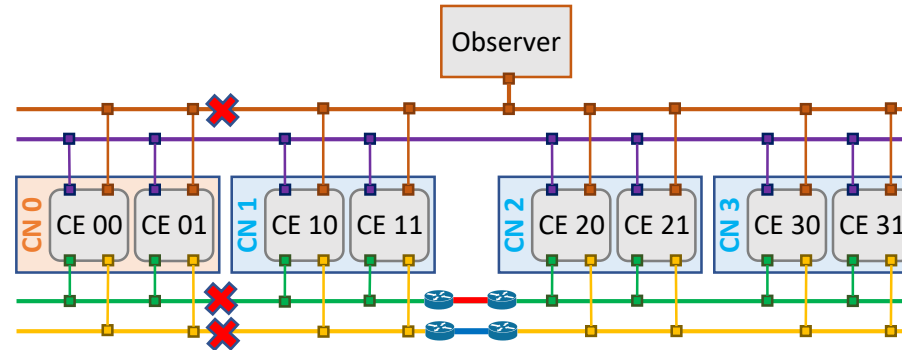
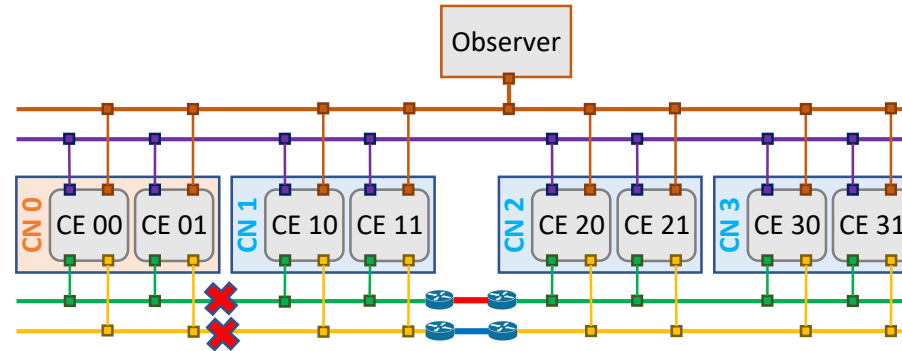
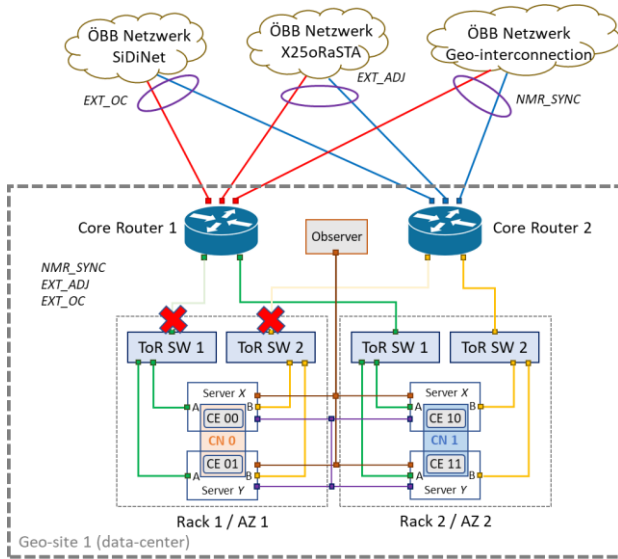
- CN0 remains master, will not see other slaves
- CN1-CN3: MEP, MTP → Not_OK, Sv → OK, new master elected. **Split brain situation**
- **CN0 has old state, new master will have another state**
 - **Issue when reintegrating – CN0 might become master with old state**
 - Only CN0 restart or shutdown of the entire Rack/AZ to reintegrate

CN0 Slave (no MCI), CN1 as master:

- CN1 remains master and will advance
- Other slaves remains slaves
- CN0 – MEP, MTP → Not_OK, Sv → OK, **Split brain**, will have a new state but not advancing (no comm)
 - **Reintegration – CN0 might become master, loss of previous state**
 - Only CN0 restart or shutdown of the entire Rack/AZ to reintegrate

CE can be launched (deployed, started) only if the infracture is connected – both NMR and service channel.

Double link failure - CN0 as master 1/2



CN0 Master (No MCI):

- CN0 remains master, will not see other slaves
- CN1-CN3: MEP, MTP → Not_OK, Sv → Not_OK, **panic** → New master NOT elected (because Observer see it)
- If G&Y for EXT → No comm. with ext, **availability issue**
 - Reintegration – CN0 maintains the state
- If EXT independent → all OK
 - Reintegration – CN0 maintains the state

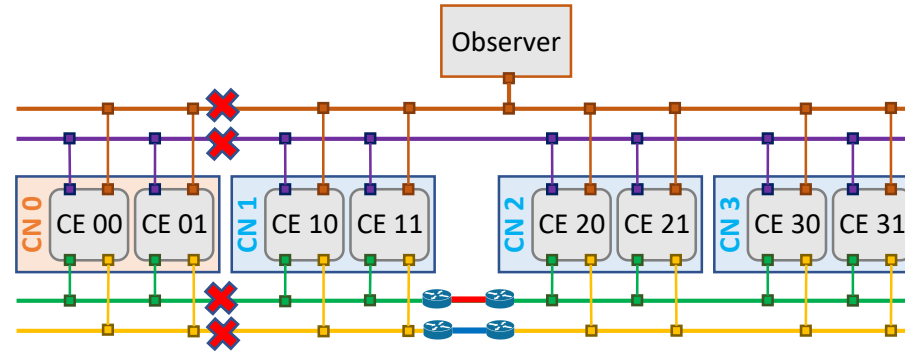
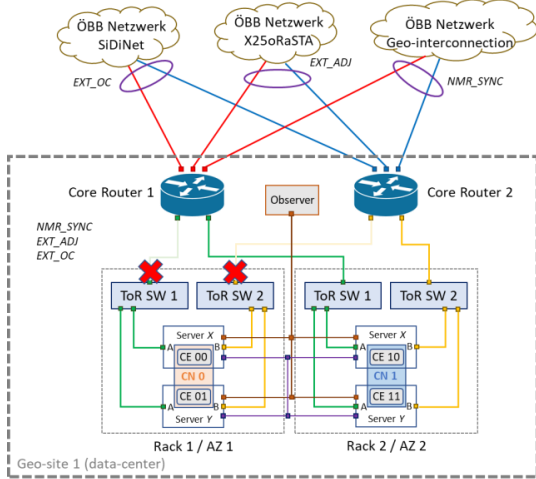
CN0 Master (No MCI):

- CN0 remains master, will not see other slaves
- CN1-CN3: MEP, MTP → Not_OK, Sv → Not_OK, **panic** → New master elected (because Observer does not see it). **Split brain situation, old state lost**
- If G&Y for EXT → No comm. with ext, availability issue
 - Reintegration – state on old master remains, new master will have another state
 - Only CN0 shutdown or panic could help to reintegrate

CN0 Master (No MCI):

- CN0 remains master, will not see other slaves
- CN1-CN3: MEP, MTP → Not_OK, Sv → OK → New master elected. **Split brain situation**
- If G&Y for EXT → No comm. with ext, availability issue
 - Reintegration – state on old master remains, new master will have another state
 - Only CN0 shutdown or panic could help to reintegrate

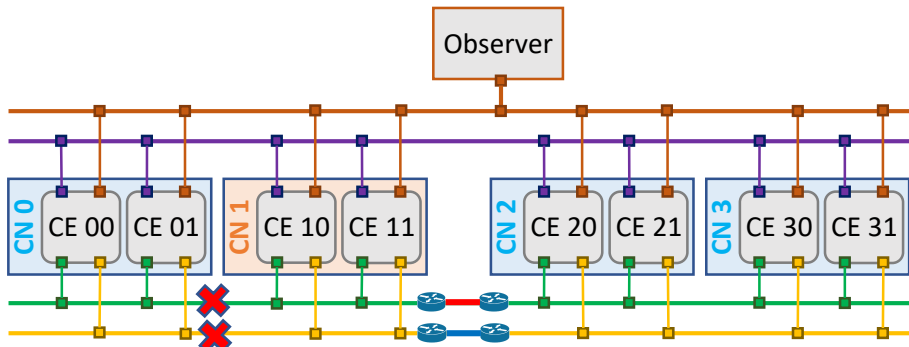
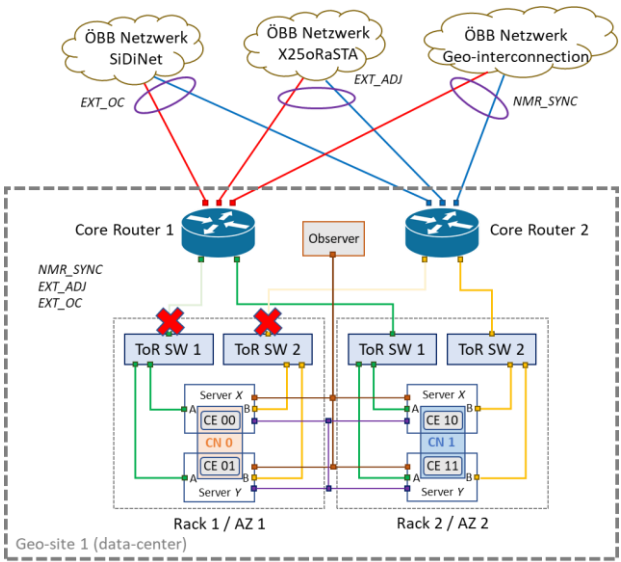
Double link failure - CN0 as master 2/2



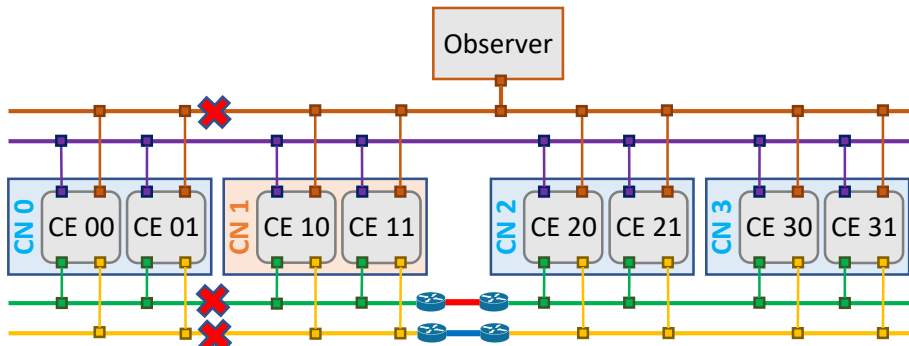
CN0 Master (No MCI):

- CN0 remains master, will not see other slaves
- CN1-CN3: MEP, MTP → Not_OK, Sv → OK, new master elected. **Split brain situation**
- If CN0 no ext comm: **CN0 has old state, new master will have new state. Issue when reintegrating.**
 - Only CN0 shutdown or panic could help to reintegrate

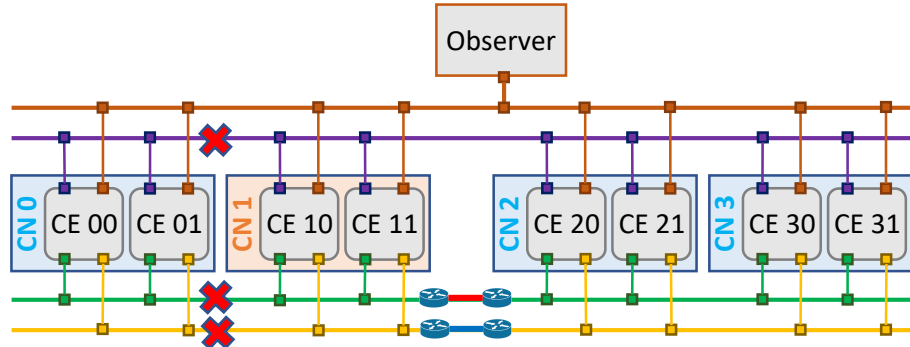
Double link failure - CN0 as slave 1/2



- CN0 Slave (No MCI):
- CN1 remains master
 - Other slaves remains slaves
 - CN0 – MEP, MTP → Not_OK, Sv → Not_OK, panic → master already exists → **recovery**

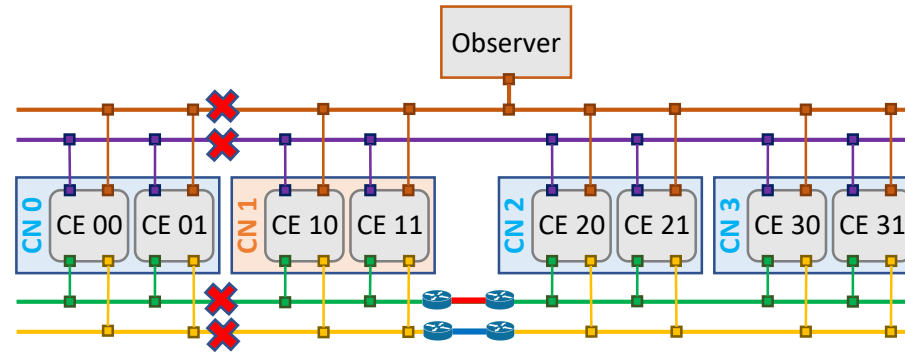
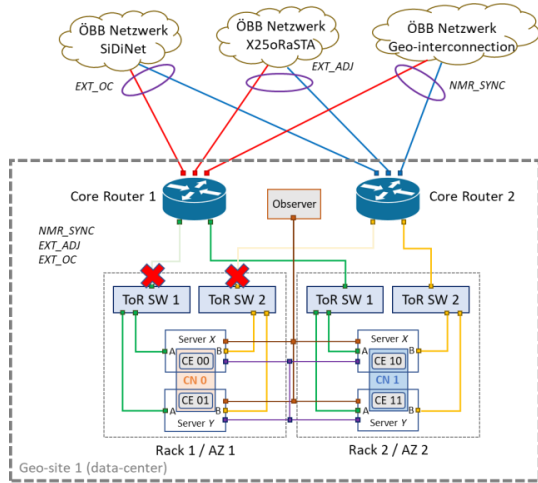


- CN0 Slave (No MCI):
- CN1 remains master
 - Other slaves remains slaves
 - CN0 – MEP, MTP → Not_OK, Sv → Not_OK, panic → master already exists → **recovery**



- CN0 Slave (No MCI):
- CN1 remains master and will advance
 - Other slaves remains slaves
 - CN0 – MEP, MTP → Not_OK, Sv → OK, Split brain, will have new state but not advancing (no comm)
 - **Reintegration – CN0 might become master, loss of previous state**

Double link failure - CN0 as slave 2/2



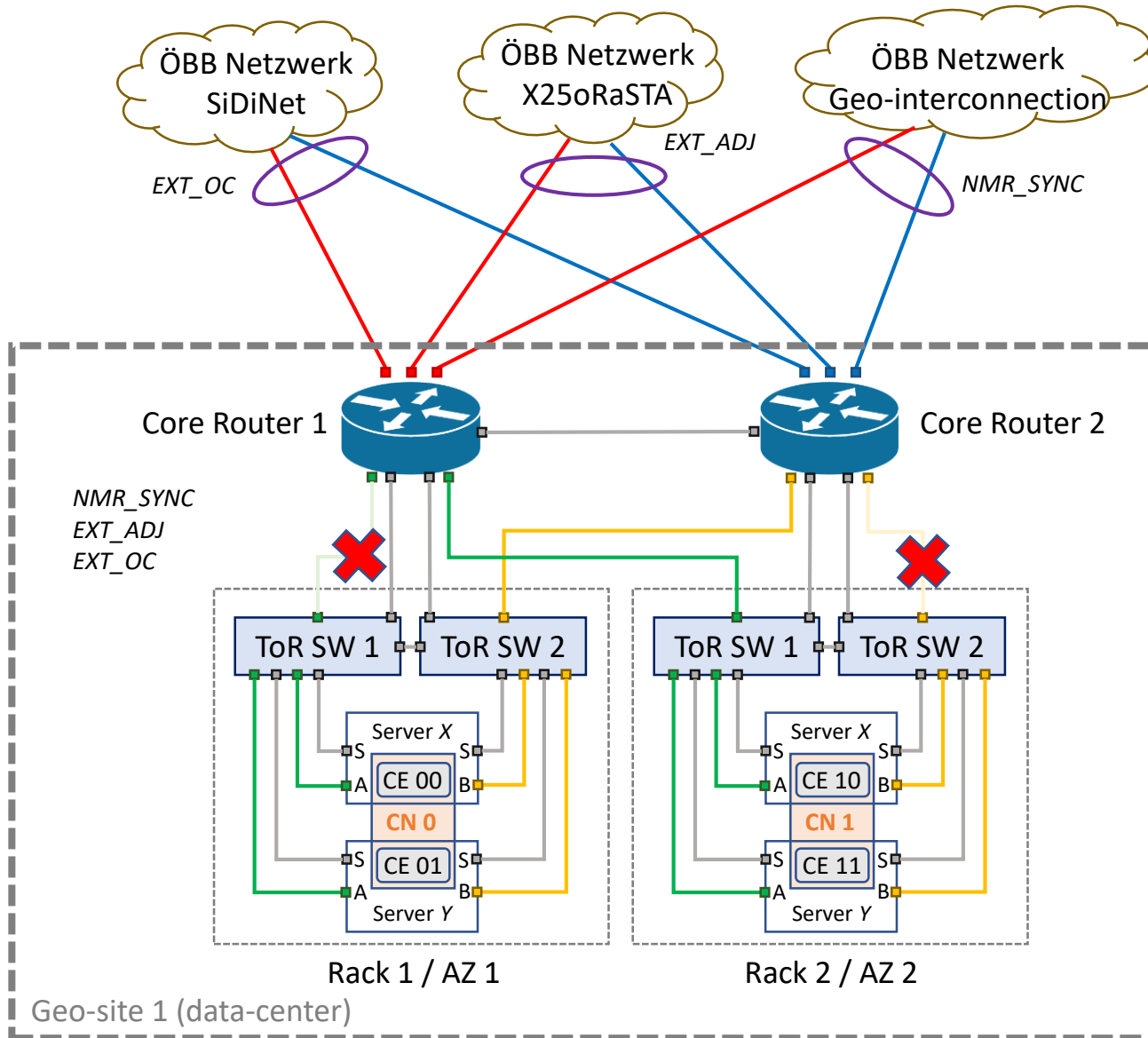
CN0 Slave (No MCI):

- CN1 remains master and will advance
- Other slaves remains slaves
- CN0 – MEP, MTP → Not_OK, Sv → OK, Split brain, will have new state but not advancing (no comm)
 - Reintegration – CN0 might become master, loss of previous state
 - Only CN0 shutdown or panic could help to reintegrate

Double link failure from cabinet to core switches

- 3 physical independent channels (3x ToR SW)
 - 3rd link (both Sv and Observer unaffected) – can handle such failures
 - If 3rd link failure
 - If some CN is in startup: will remain in recovery until the link is established
 - If new master election process (master CN is down)
 - Either double link failure, i.e. we have already 3 failures
 - Or master has failure – if inside failing cabinet – CN will end in recovery, other redundancies (in other cabinets) will remain state and elect new master.
- 2 independent physical channels (2x ToR SW)
 - System will function in double link failure but reintegration will be a hazard – all masters in the affected must be restarted

Network topology for split-brain analysis 2x ToR SW with bonded service channel



4x2oo2

• CN0 Master:

- CN0 remains master and will see CN2 and CN3, will not see CN1
- CN1 will panic, on start will go to the recovery state (Observer won't allow to become master)
- CN2 and CN3 (other geo-site): no effect

• CN1 is Master: The same as for CN0

• CN0 and CN1 are Slaves: no effect

- No hazards, no split-brain
- Reintegration – no issues

2x2oo2

• CN0 Master:

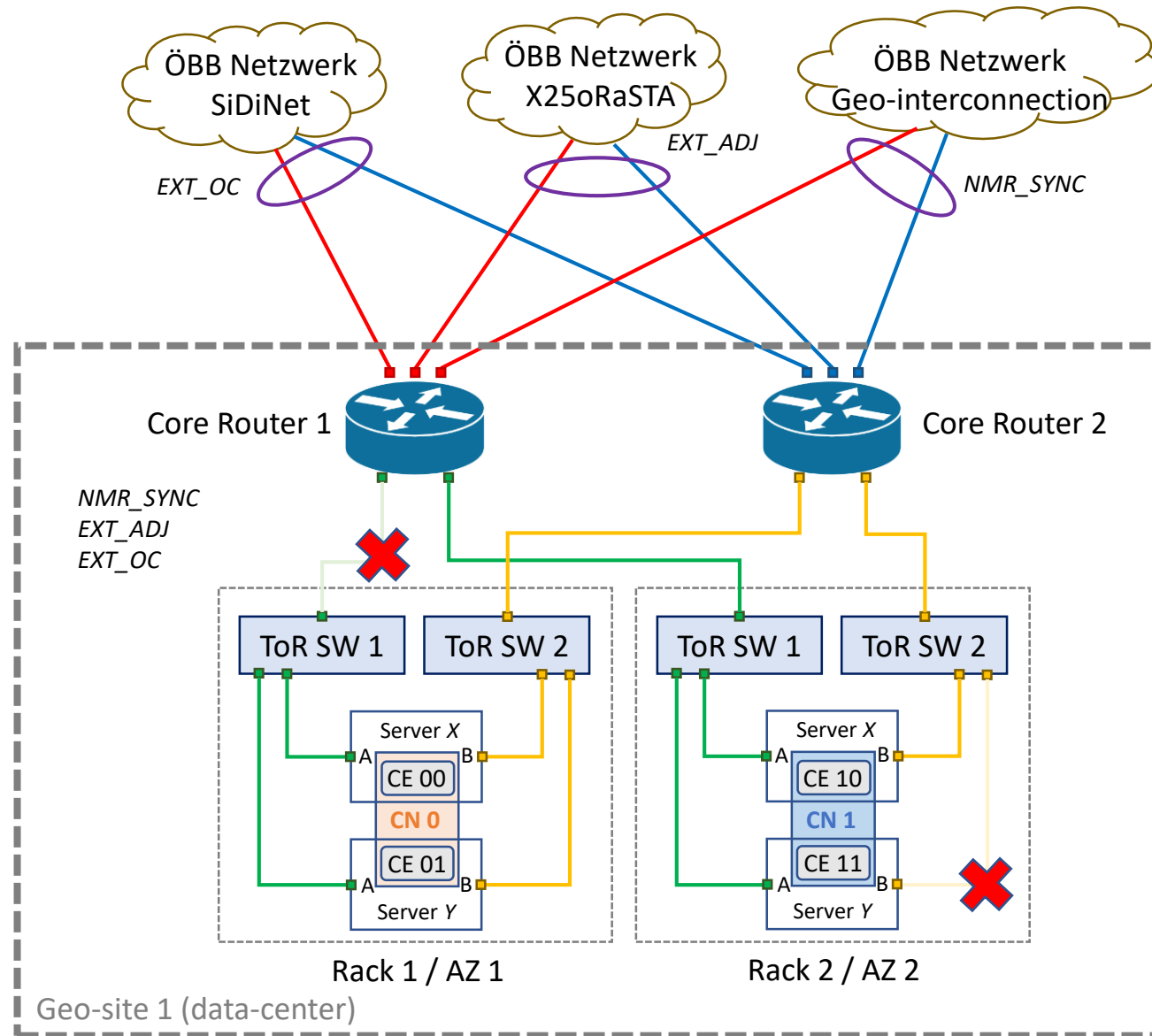
- CN0 remains master
- CN1 will panic, on start will go to the recovery state (Observer won't allow to become master)

• CN1 is Master:

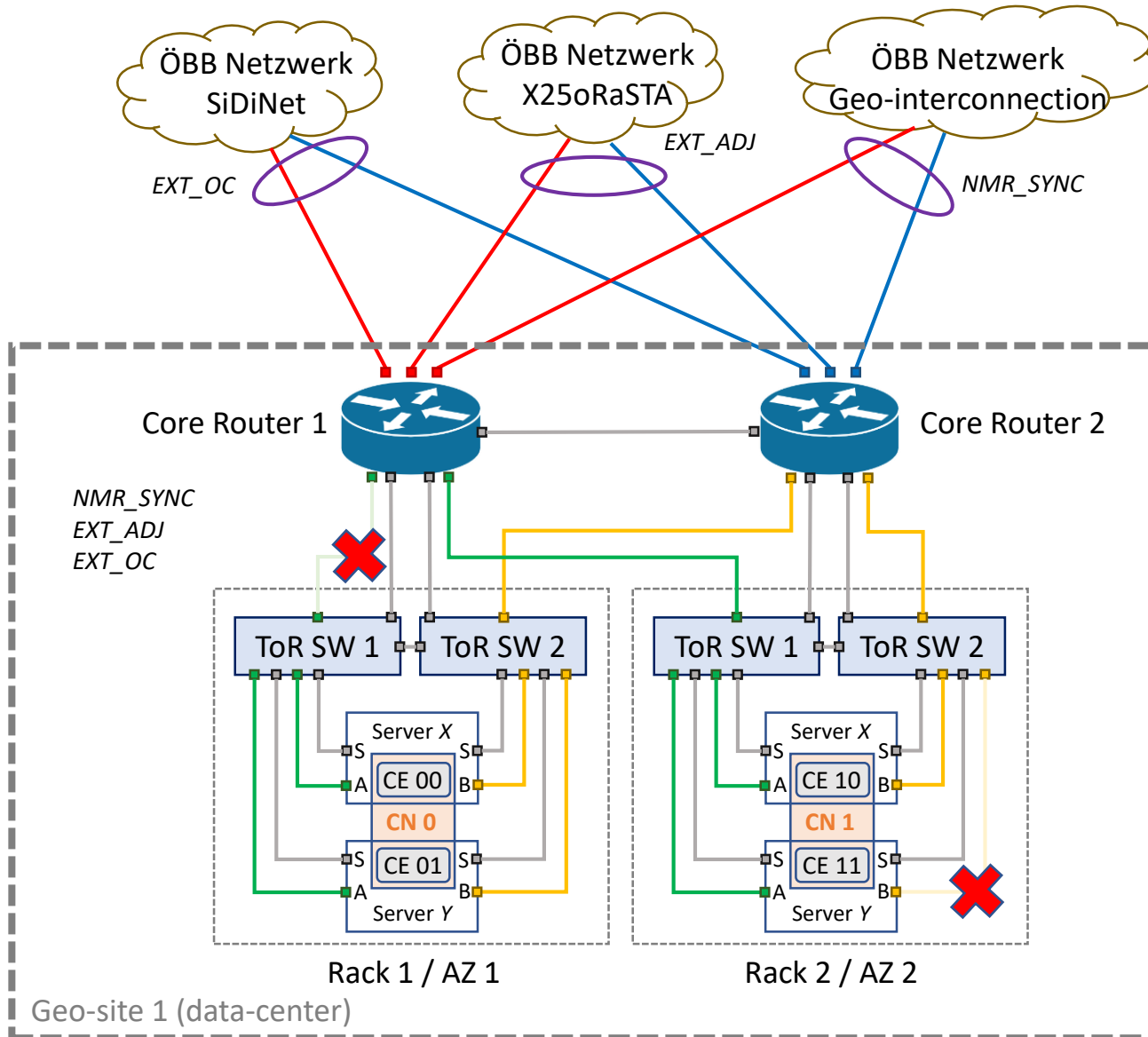
- CN1 Remains master
- CN0 will panic and goes to recovery (Observer won't allow to become master)

- No hazards, no split-brain
- Reintegration – no issues

Double failure between and inside racks



Network topology for split-brain analysis 2x ToR SW with bonded service channel



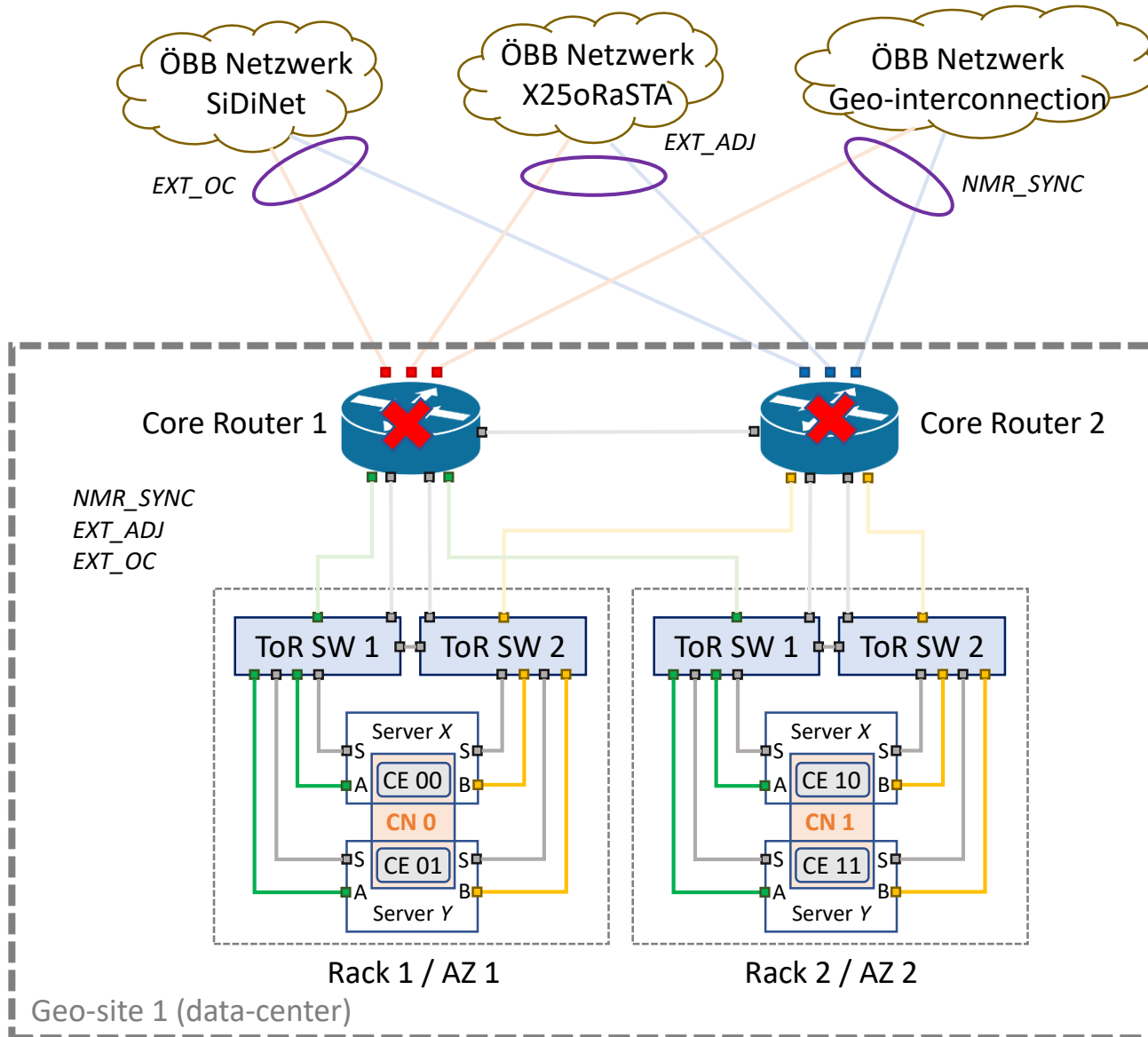
4x2oo2

- **CN0 Master:**
 - CN0 remains master and will see CN2 and CN3, will not see CN1
 - CN1 will panic, on start will go to the recovery state (Observer won't allow to become master)
 - CN2 and CN3 (other geo-site): no effect
- **CN1 is Master:** CN0 will panic and goes to recovery (Observer won't allow to become master), CN2 and CN3 operates normally
- **CN0 and CN1 are Slaves:** no effect
- No hazards, no split-brain
- Reintegration – no issues

2x2oo2

- **CN0 Master:**
 - CN1 will panic, on start will go to the recovery state (Observer won't allow to become master)
- **CN1 is Master:**
 - CN1 Remains master
 - CN0 will panic and goes to recovery (Observer won't allow to become master)
- No hazards, no split-brain
- Reintegration – no issues

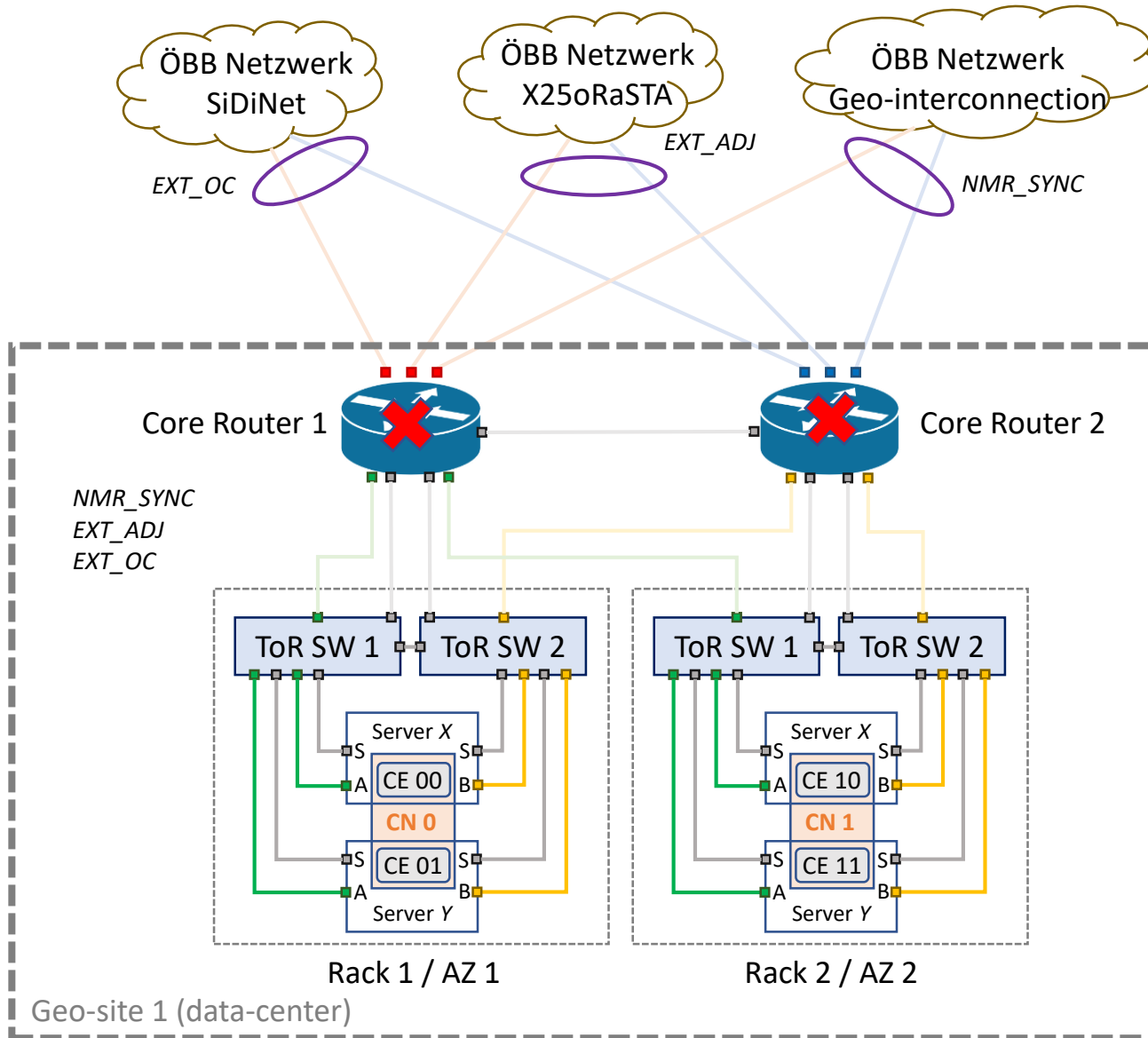
Network topology for split-brain analysis 2x ToR SW with bonded service channel



2x2oo2

- CN0 Remains master, will not see CN1
- CN0 will not advance any computation because no communication with external networks
- CN1 – master timeout, MEP, MTP → Not_OK, Sv → OK, Split brain, CN1 will not advance because no communication with external networks
- **External networks:** No communication with any external network
- **Split-brain situation but no communication with EXT and advancing internal state, overall availability issue**
- **Reintegration (Router 1 is operational):**
 - Dual master detected
 - CN0 remains master, CN1 will panic and recover as slave CN
 - No conflicting state information

Network topology for split-brain analysis 2x ToR SW with bonded service channel

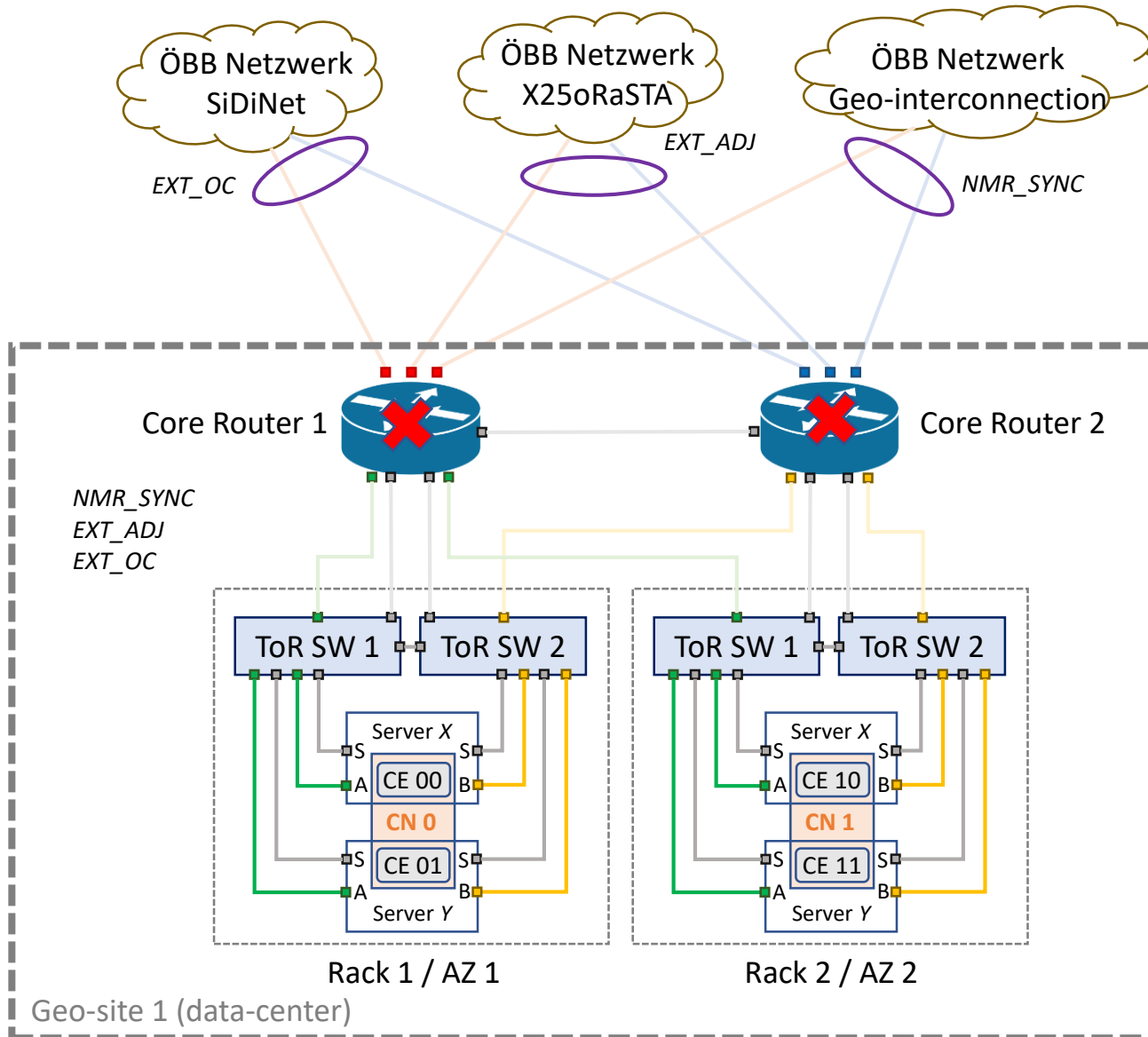


4x2oo2

CN0 Master:

- CN0 Remains master, will not see CN1, CN2, CN3
- CN0 will not advance any computation because no communication with external networks
- CN1 – master timeout, MEP, MTP → Not_OK, Sv → OK, Split brain, CN1 will not advance because no communication with external networks
- CN2 and CN3: one will become master (master timeout, MEP, MTP → Not_OK, Sv → OK)
 - Will advance state because has connectivity with EXT
- **Observer:** not used and needed, only later if geo-site 2 will go fully down and will need to start
- **3 CN masters – safe because only one master is connected to EXT, other 2 masters are fully isolated**
- **Reintegration:**
 - **Multiple masters detected**
 - **CN0 remains master, CN1, CN2 (or 3) will panic and recover as slave CN**
 - **Hazard: Loss of state because CN0 has old state and CN2/3 already advanced**
 - **Safe reintegration procedure:**
 - Shutdown of all CN on geo-site 1, reconnect core routers, start CNs – no impact on availability

Network topology for split-brain analysis 2x ToR SW with bonded service channel

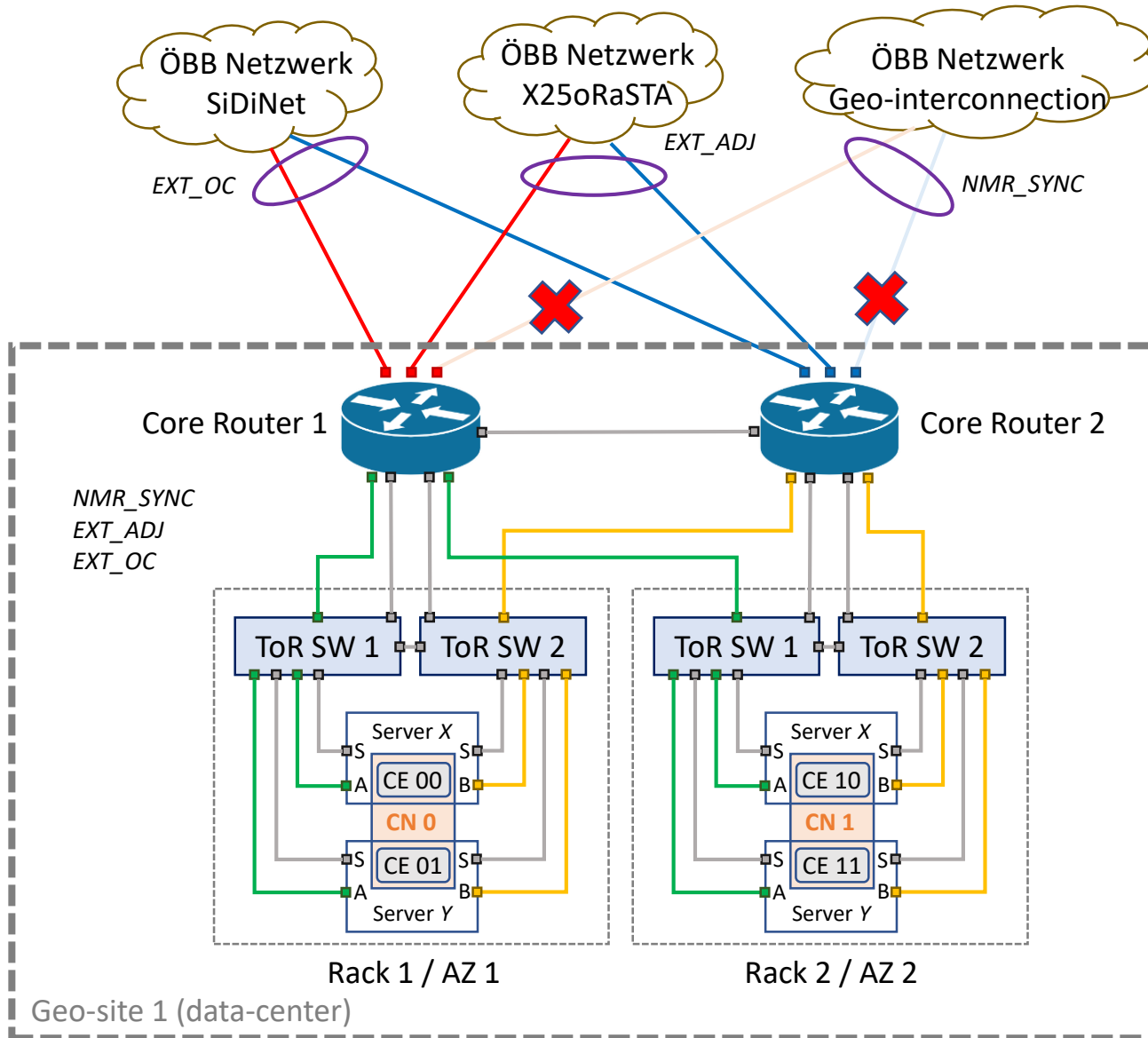


4x2oo2

CN0 and CN1 Slaves:

- Master in geo-site 2 remains master, geo-site 2 will continue normal operation but will not see CN1 and CN2
- CN0/1 – master timeout, MEP, MTP → Not_OK, Sv → OK, Split brain, CN will not advance because no communication with external networks
 - One CN will be master and another slave
- Observer:** not used and needed, only later if geo-site 2 will go fully down and will need to start
- 2 CN masters – safe because only master at site 2 is connected to EXT, master in site 1 is fully isolated**
- Reintegration:**
 - Multiple masters detected**
 - One master will panic, another will remain master**
 - Hazard: Loss of state because previously isolated master has old state**
- Safe reintegration procedure:**
 - Shutdown of all CN on geo-site 1, reconnect core routers, start CNs – no impact on availability

Network topology for split-brain analysis 2x ToR SW with bonded service channel



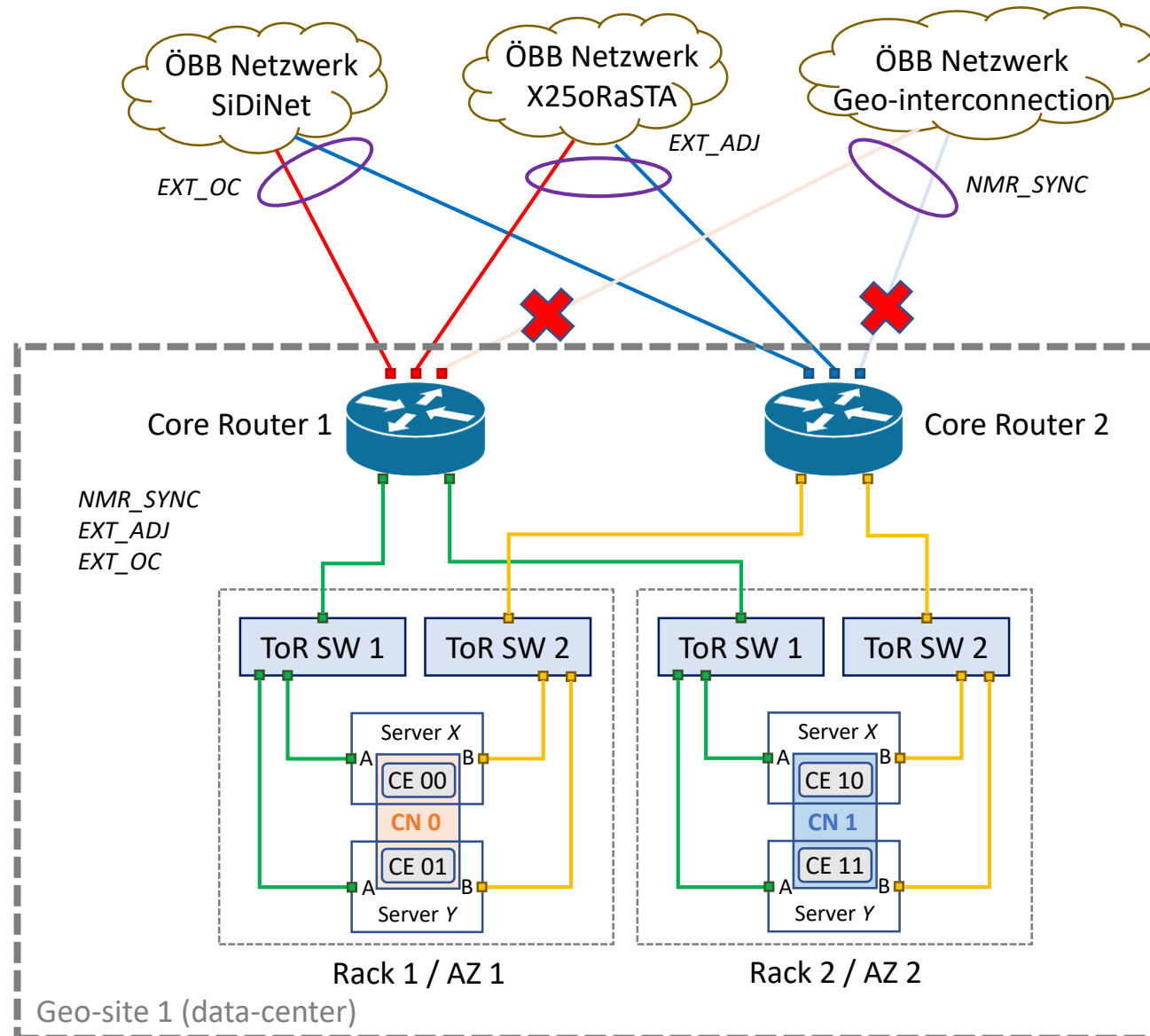
4x2oo2

CNs in different geo-sites cannot see each other

Master in site 1 (CN0 or CN1):

- CN0 remains master, CN1 remains slave
- CN0 master will operate normally (but will see CN2 and 3 as UNKNOW)
- CN1 as slave will operate normally
- **CN2/3:** master timeout, MEP, MTP → Not_OK, Sv → OK
- Either CN2 or CN3 will become master and continue operation
- **Safety hazard: Split-brain: 2 active masters, having active communication with OCs and ADJ systems**
 - **Reintegration: One CN master will panic, another one will continue operation. State on both CN might diverge and be inconsistent**

Double failure in geo-interconnect network



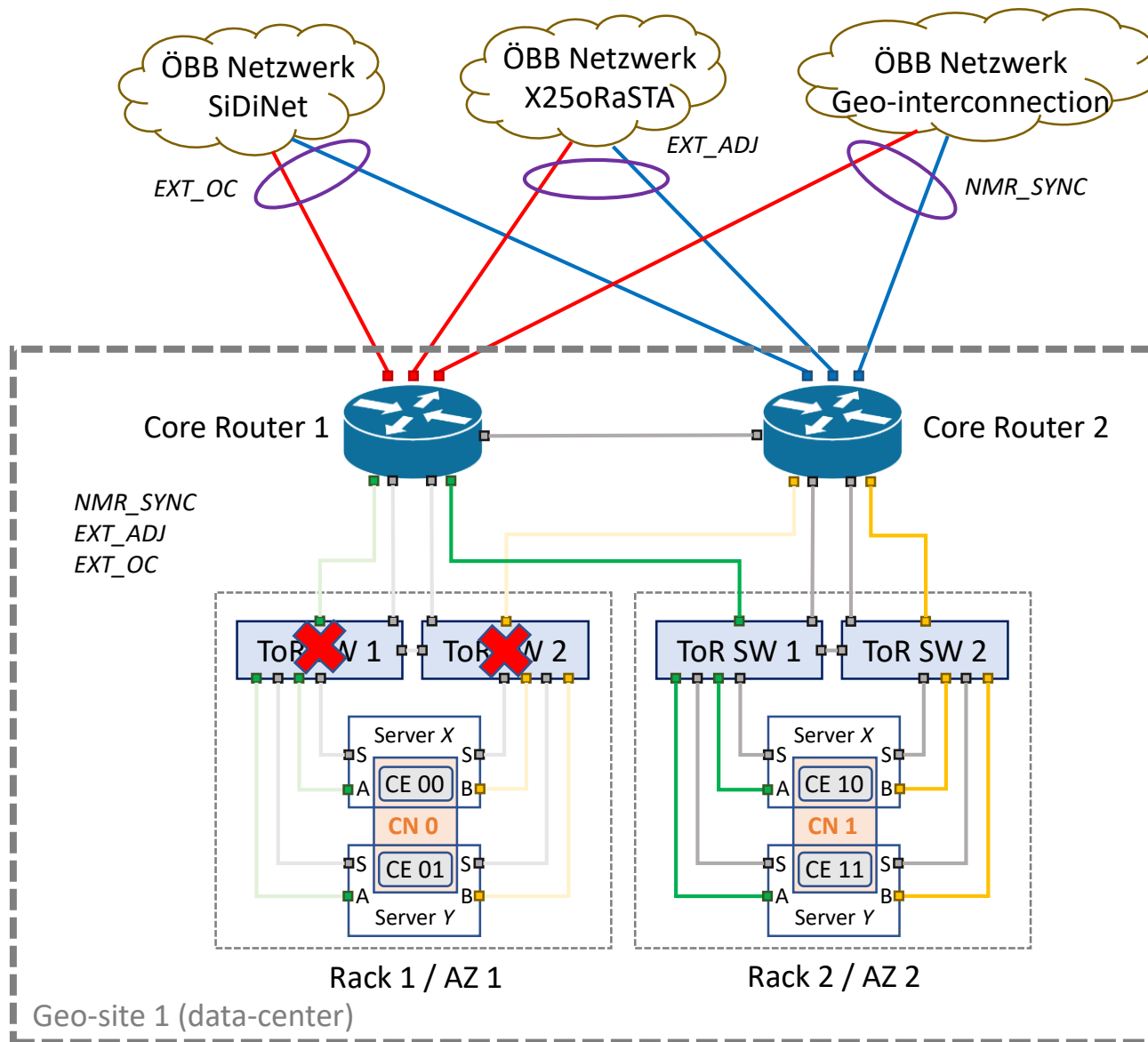
2x2oo2: not applicable

4x2oo2:

- **CN effect:** All 4 CNs are operational
- **CG effect (default):**
 - CNs in different geo-location cannot see each other
 - Location with Master CN – remains master
 - Location with Slave CN – Starts MEP
 - MTP cannot decide
 - MCI queried
- **External networks:** Site with master CN continues communication with all EXT. Site with slaves will not be operational
- **No risk of split brain**
 - No safety hazard because site with master continues and site with slaves is unable to elect the master
- **No availability issue**
 - Site with master is connected to all external networks

Double ToR SW failures

Network topology for split-brain analysis 2x ToR SW with bonded service link



Master CN (on the left):

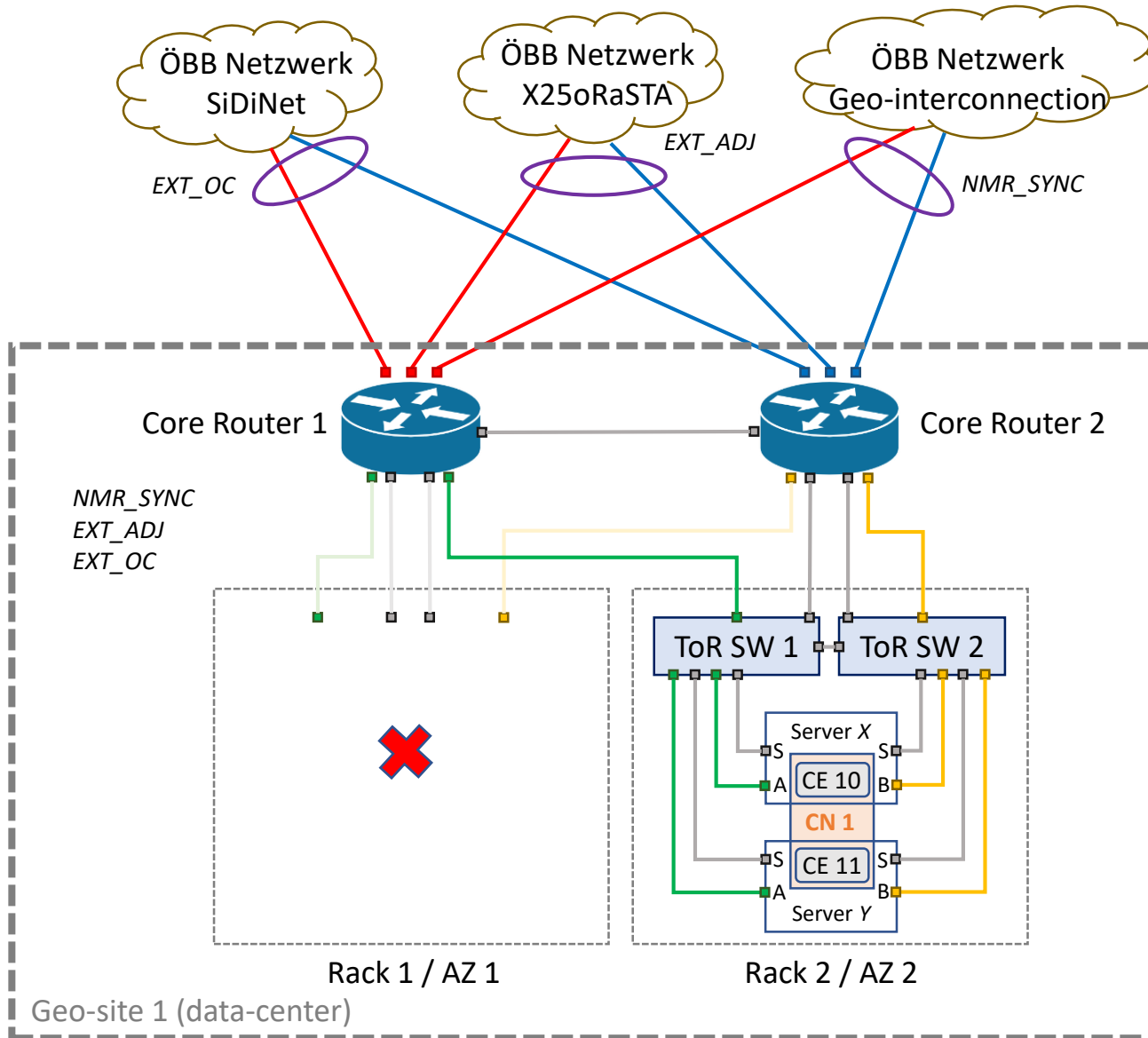
- CN0 goes to panic, CN will not start
- CN1-CN3: Master tick timeout → MEP, MTP → Not_OK, **Sv → OK**, New master elected, **no loss of the state**
- **No hazards, no split-brain, no disruptions**
- **Reintegration – CN0 recovers**

Slave CN (on the left):

- CN0 goes to panic, CN will not start
- **No hazards, no split-brain**
- **Reintegration – CN0 recovers**

Power outages

Network topology for split-brain analysis 2x ToR SW with bonded service channel



Availability zone is gone

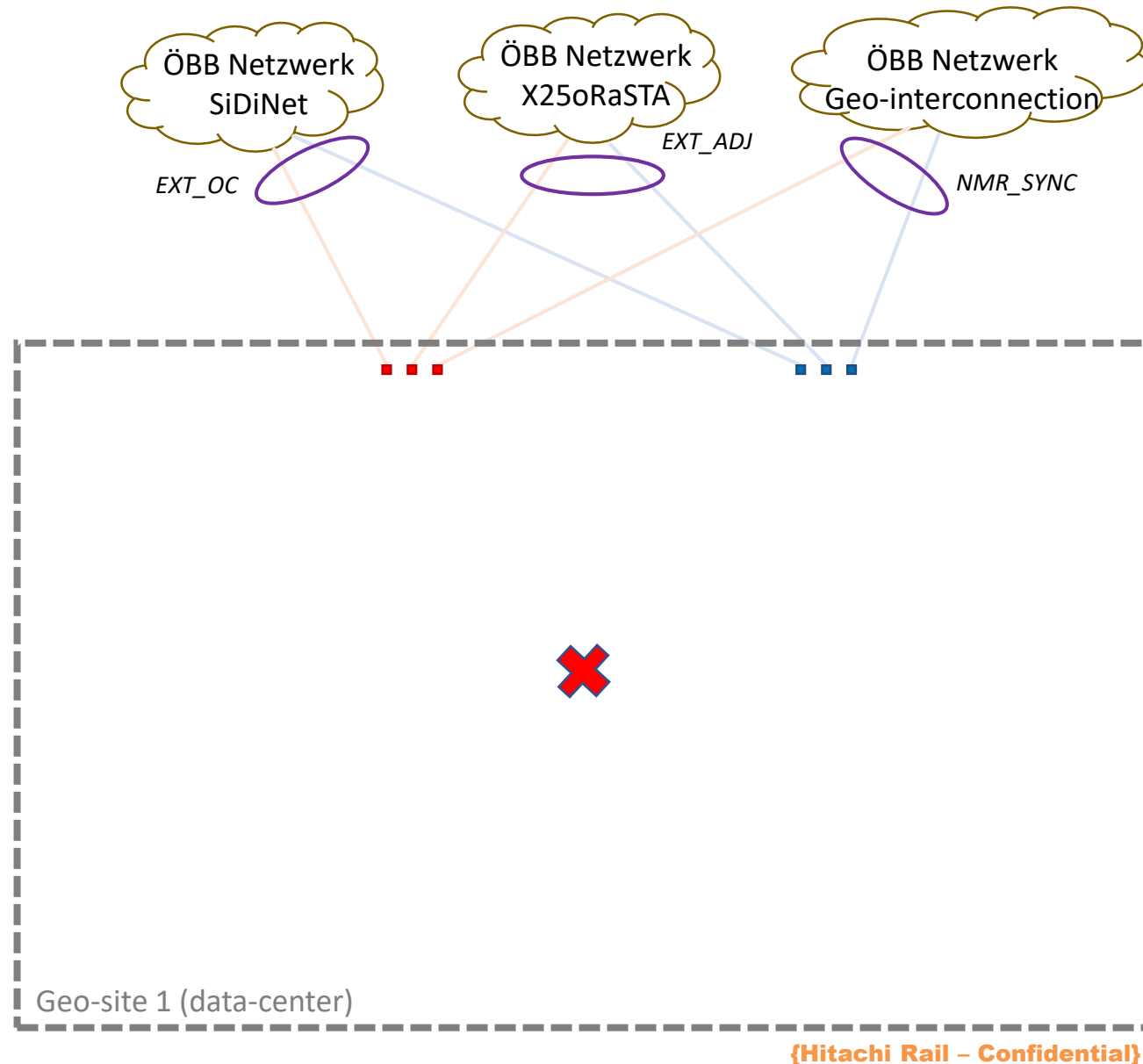
Master CN (on the left):

- CN1/2/3: Master tick timeout → MEP, MTP → Not_OK, Sv → OK, New master elected, no loss of the state
- No hazards, no split-brain, no disruptions
- Reintegration – CNO recovers

Slave CN (on the left):

- CN1/2/3: continues operation, one of them is master
- No hazards, no split-brain
- Reintegration – CNO recovers

Power outage of the entire geo-site



2x2oo2

- The whole VIL is down, no operations
- No risk of split-brain
- **No risk of split brain**

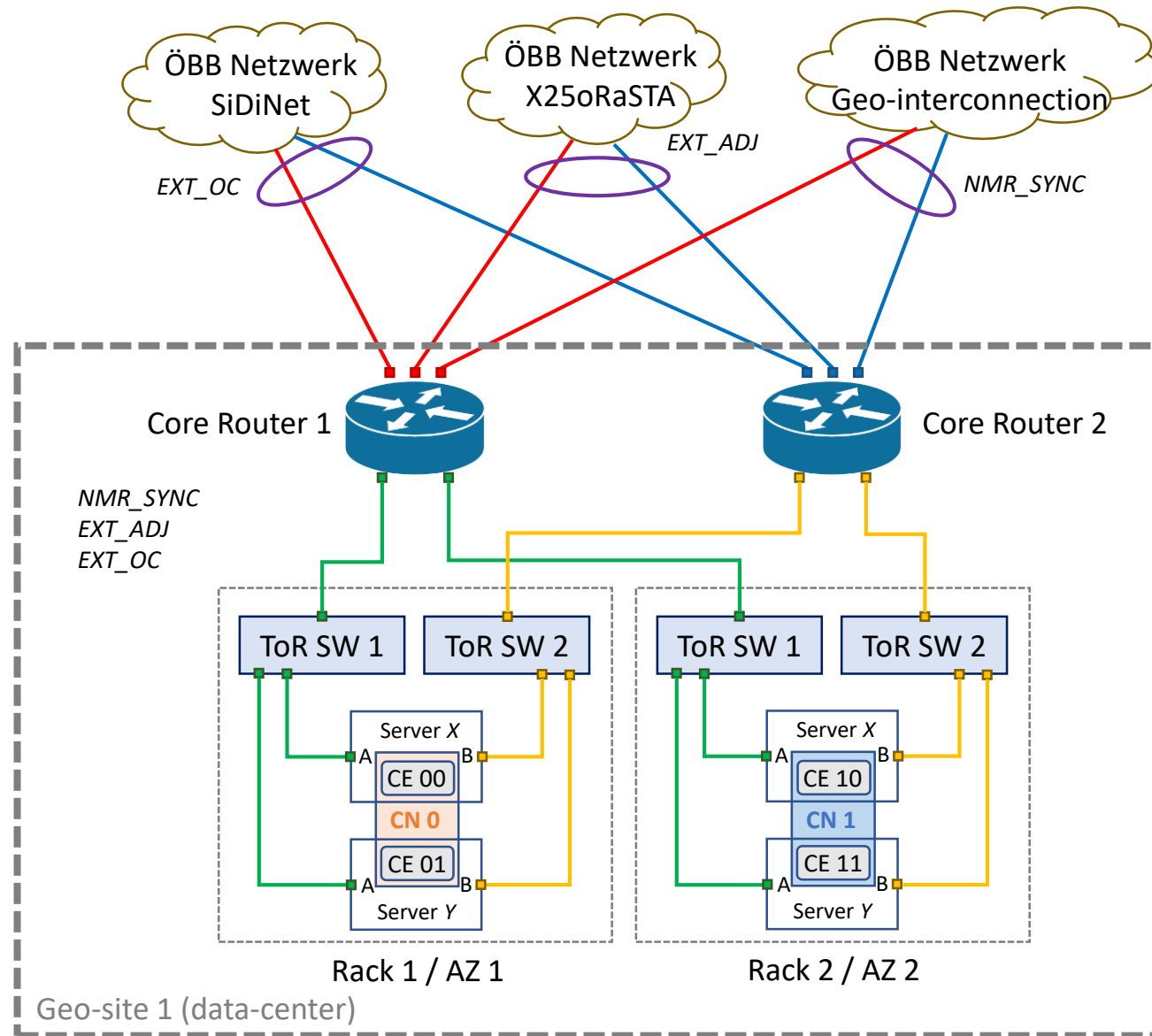
4x2oo2

CN2/3 in site 2:

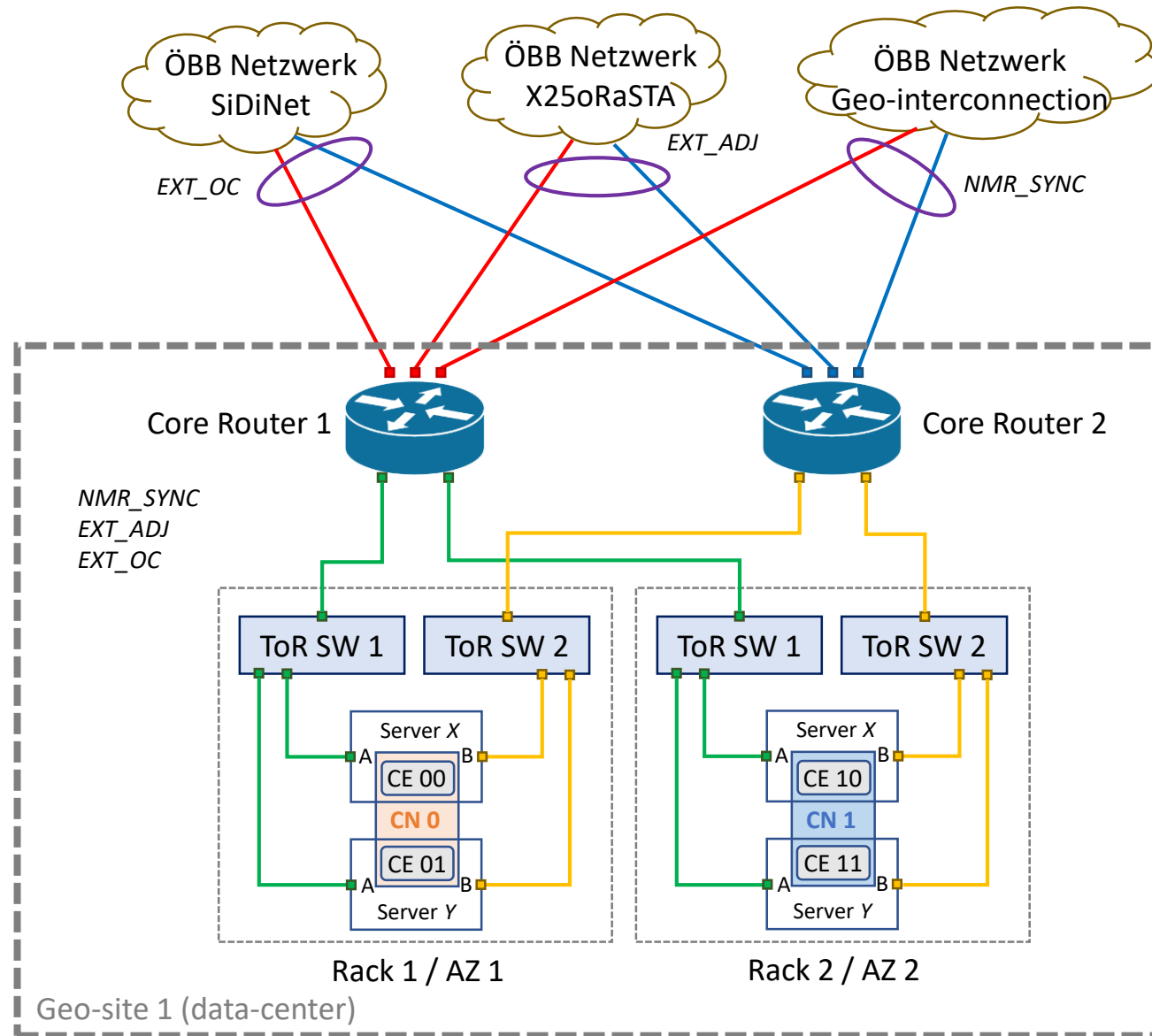
- Master tick timeout → MEP, MTP → Not_OK, **Sv → OK**, New master elected, **no loss of the state**
- **No hazards, no split-brain, no disruptions**
- **Reintegration – CN0/1 recover**

Backup slides

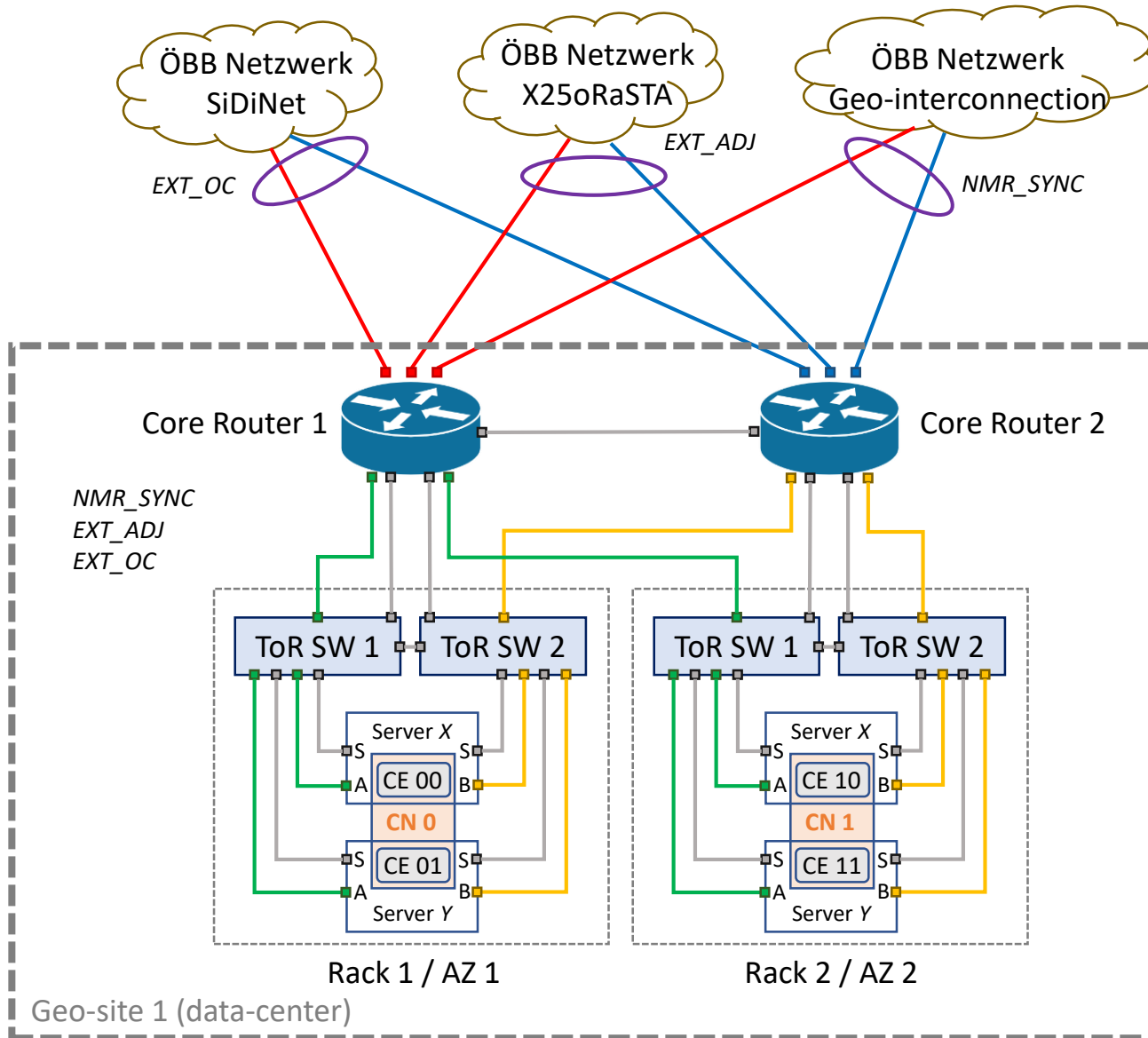
Network topology for split-brain analysis



Network topology for split-brain analysis



Network topology for split-brain analysis 2x ToR SW with bonded service channel



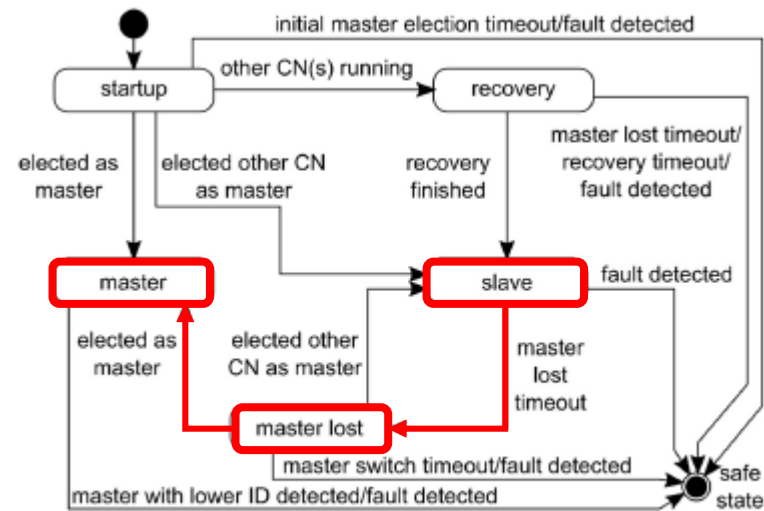


Figure 3 NMR state machine of a CN

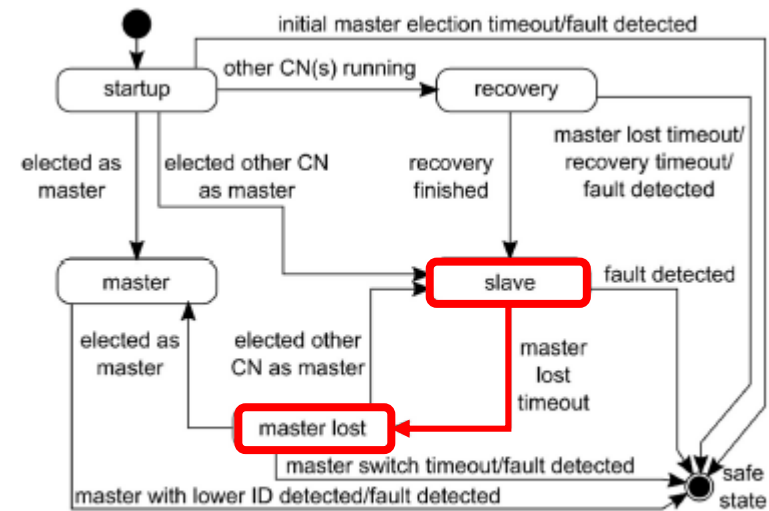
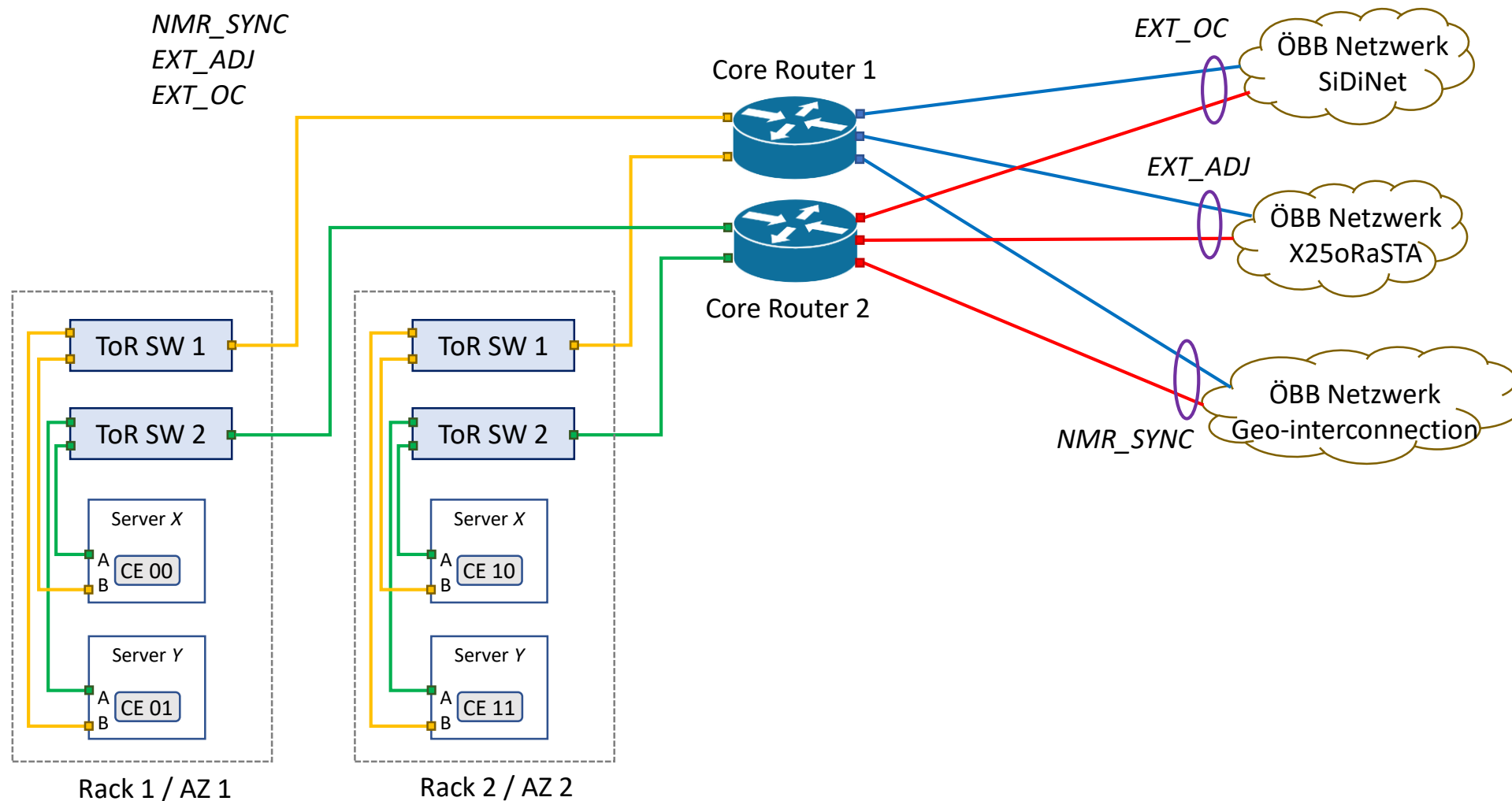
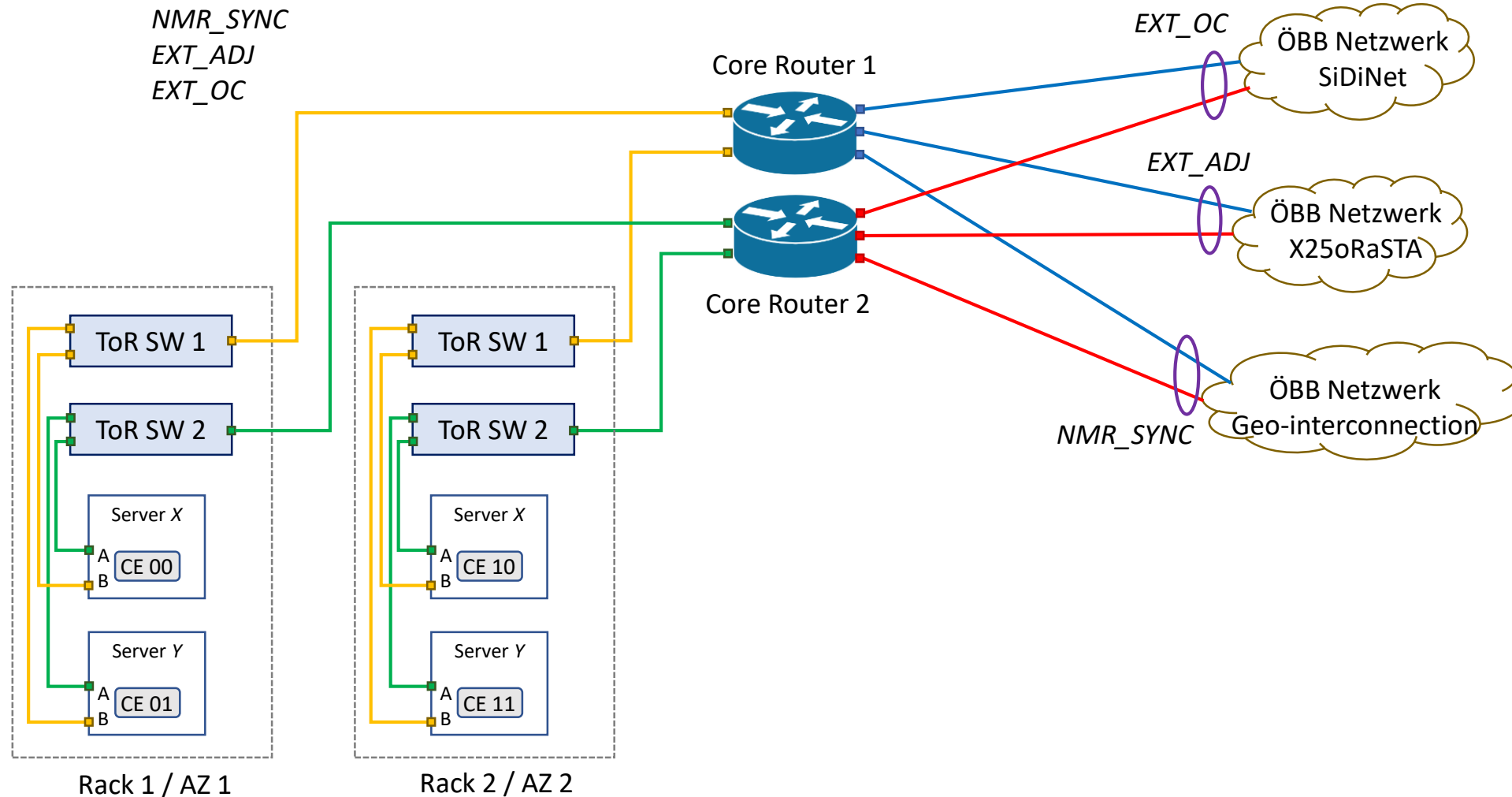


Figure 3 NMR state machine of a CN

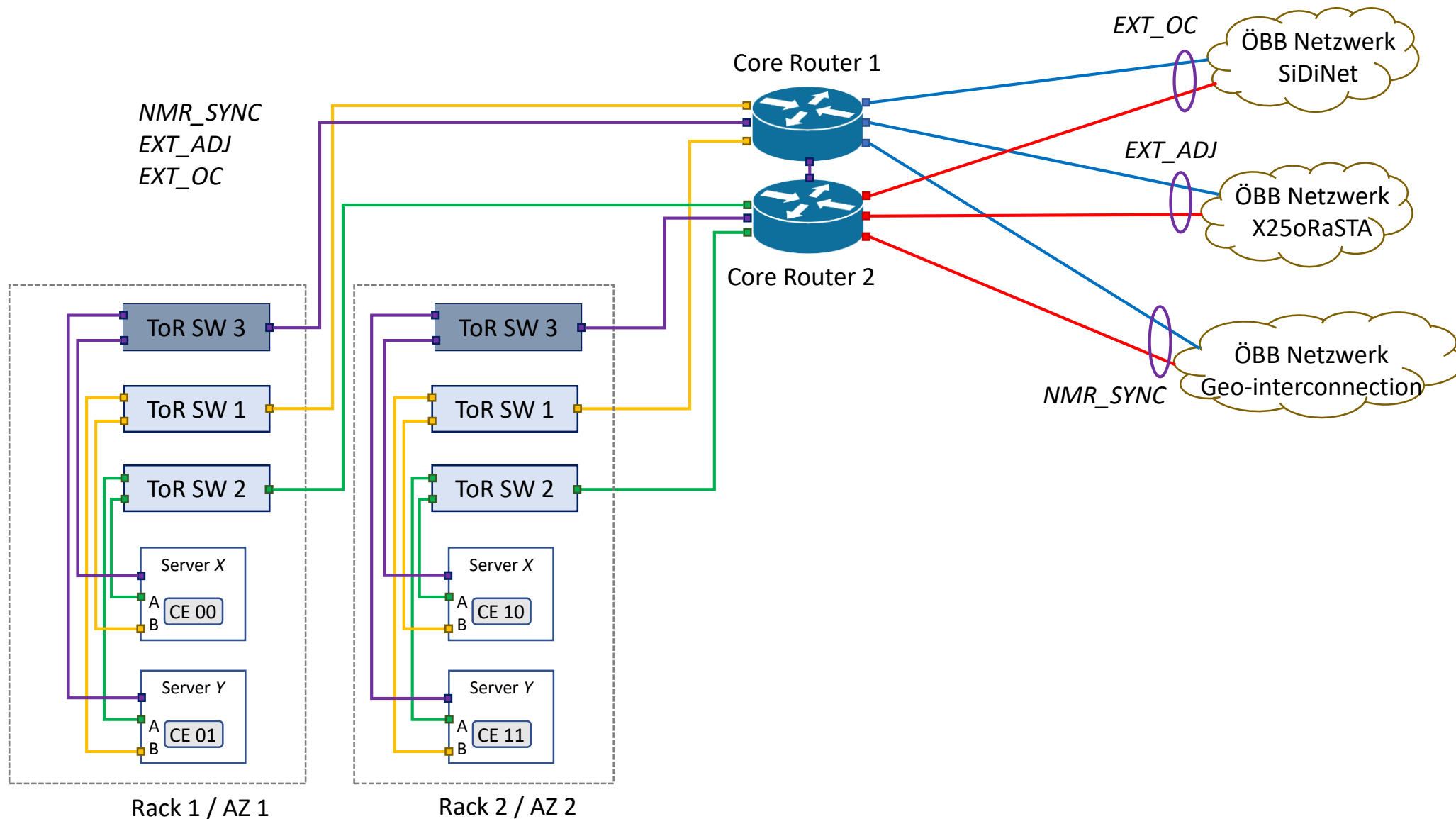
Network topology for split-brain analysis



Network topology for split-brain analysis



Network topology for split-brain analysis



Double failure between racks and core routers

